EIGSEP Memo 005

Marjum Pass 2026-07: beam reconstruction from the transmitter raster

EIGSEP Team

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Abstract

How box-air's beam is measured from the 07-17 transmitter raster, and how well. The transmitter's comb teeth, seen through box-air as it rolls and tips, sample the beam's response to two orthogonal polarizations over the whole sphere of directions. This memo describes the data selection, the background under each tooth, the beam model (spherical harmonics times a smooth spectral basis, regularized toward the HFSS simulation), the geometry fit and the held-out test, and states how far the reconstructed beam improves on HFSS. The reference product is `derived/beam/empirical_raster_v0013`, fitted on geometry release `v0004` (memo 002). This memo does not refit it: every number, figure and table is computed by `make_products.py` in this directory from the product's per-sample data and predictions (Appendix A). Pointing, the mount and the azimuth zero are in memo 003.

1 Summary

- **Data**: 64 transmitter teeth between 64.2 and 240.5 MHz (25 on arm 0, 39 on arm 1), 6845 samples over 20:26:00–21:28:40 UTC on 07-17, split by 12° azimuth stripes into training (4497), validation (1297) and test (1051) samples.
- **The empirical beam beats HFSS on the test stripes**: fractional rms 0.083 against 0.111 (means over teeth weighted by sample count), better on 59 of 64 teeth, with the worst tooth at 0.173 against 0.321. The gain is largest above 190 MHz, where HFSS is worst (Table 4).
- **The residual is structure, not noise**: it is ∼6× the white-noise floor (0.014). What is left is in the model, the background or the geometry, and is largest where the transmitter is 60–140° from the boresight.
- **Geometry**: the antenna and transmitter are geometry release `v0004` (memo 002). Against the previous product, fitted on `v0001`, the transmitter's direction moves by 0.38°; the fitted elevation zero absorbs it (+0.56° to +0.93°) and the test scores do not change at the third decimal. The fit adopts memo 003's convention and finds the same azimuth zero (−16°) and a transmitter arm 0 at bearing 84°, along the axis photographed as rx1. Rerun with the opposite azimuth handedness, the HFSS beam fits worse (test 0.093 against 0.088), which agrees with memo 003.
- **Limits**: power-only data leave the beam's phase undetermined; directions the raster never sampled (44% of HEALPix pixels are sampled) are set by the HFSS prior; the transmitter offsets sit at their ±1.06 m bound, which v0004 does not supply (§4).

2 Data

Samples. Box-air, antenna-input rows of the elevation-sweep raster, 20:26:00–21:28:40 UTC on 07-17 (memo 003, The 07-17 transmitter raster), with pointing-table v2.0 quality `ok`: commanded azimuth and IMU elevation. Isolated elevation-solution glitches are dropped. Throughout the raster box-air rolls about its boresight in 5° steps while the elevation axis turns through full revolutions, so the transmitter is seen at every angle from the boresight and at a range of roll angles.

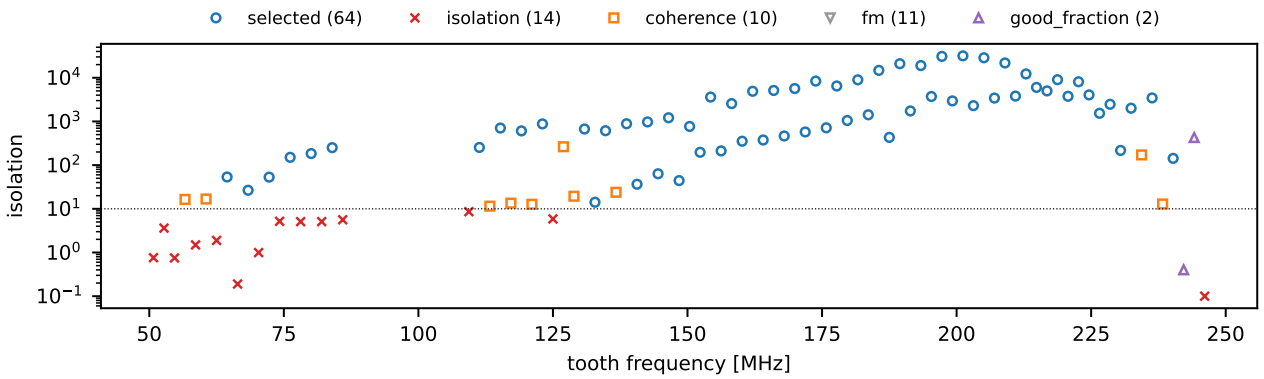


Figure 1: Tooth isolation against frequency, marked by the reason each candidate was kept or dropped. The dotted line is the isolation threshold.

Background under each tooth. The comb sits on box-air's sky and receiver spectrum. For each integration a robust DPSS model is fitted to the spectrum in 48 MHz windows with a 150 ns delay half-width, masking each tooth's channel and the channels either side, and the FM band (87–108.5 MHz), with 4σ clipping; the tooth signal is the measured power minus this background. Since v0012 the background keeps every DPSS mode with concentration $\geq 10^{-6}$ (21 modes); the earlier floor$(2NW)+1$ rule could not represent a constant near the window edges (`eigsep_data.beam_mapping.tx_background`). Broadband dropouts are masked by a gap-ratio test, and `raster_flags_v0001` removes 295 samples and 32 tooth spikes.

Which teeth. The candidates are the 101 channels $\equiv 0 \bmod 8$ inside the HFSS frequency range. A tooth is kept if it is isolated (the 90th percentile of its signal-to-background ratio over the raster is at least 10 times the larger of its background error in a transmitter-off control interval, 17:03–17:54, and its local residual), if at least half its samples are good, and if it is coherent with its neighbours: its log signal must correlate at ≥ 0.9 with the mean of the two nearest selected teeth on the same transmitter arm, iterated by dropping the least coherent tooth (`tooth_selection_v0002`). 14 fail isolation, 11 are in the FM band, 10 fail coherence and 2 have too few good samples (Fig. 1). The coherence test removes teeth with intermittent RFI that pass isolation.

3 Model

Coupling. Box-air is fixed in position relative to the transmitter, so the transmitter's own gain toward box-air is the same for every sample and is absorbed, with the drive and the cable losses, into one positive gain g_f per tooth. The model power for tooth f, driven on transmitter arm $a(f)$, is

$$P_f(t) = g_f \left| \mathbf{E}_f(\theta_t, \phi_t) \cdot \hat{\mathbf{e}}_{a(f)}(t) \right|^2, \tag{1}$$

where (θ_t, ϕ_t) is the transmitter's direction in box-air's body frame (memo 003, Eq. 1), $\hat{\mathbf{e}}_a$ is the field direction of transmitter arm a (horizontal dipoles at polarization angle α and $\alpha + 90°$) expressed in the same frame, and $\mathbf{E}_f$ is box-air's complex vector far-field pattern. Both arms see the same beam through orthogonal polarizations.

The simulated beam. Dominic's HFSS model of box-air's bowtie antenna gives $\mathbf{E}_f$ as Cartesian field components on a HEALPix grid (nside 32) at 52 frequencies, 50.8–250.0 MHz (`eigsep_data/hfss_beam_maps/bowtie_beam.npz`). The fields are normalized to unit mean power at 190 MHz with the common phase removed, and interpolated to the teeth.

The empirical beam. Each Cartesian component is a sum of real spherical harmonics to $\ell_{\max} = 8$ (81 terms) with coefficients that are themselves smooth in frequency, expanded in a spectral basis: $\mathbf{E}_f(\theta, \phi) = \sum_k s_k(f) \sum_{\ell m} \mathbf{C}_{k\ell m} Y_{\ell m}(\theta, \phi)$. Two spectral bases are fitted: 8 DPSS modes with a 12.5 ns delay half-width (sinc-continued across the band), and 5 principal components of the HFSS fields. $\ell_{\max} = 8$ represents the HFSS

fields to 0.0039 worst-case relative error, and the two spectral bases to under 1%. The loss is the mean over teeth of the squared residual, each tooth scaled to unit rms, plus a prior $\lambda \|\mathbf{C}-\mathbf{C}_0\|^2/\|\mathbf{C}_0\|^2$ toward the HFSS beam's own coefficients $\mathbf{C}_0$; the gains are fitted in log so they stay positive (JAX, L-BFGS-B; `tx_fit.JointBeamFit`).

Training, validation and test. Samples are assigned by stripes of commanded azimuth 12° wide: stripe k is validation if $k \bmod 5 = 1$, test if $k \bmod 5 = 3$, and training otherwise. The beam is fitted on training samples at $\lambda = 10^{-2}, 10^{-3}, 10^{-4}$, and λ is chosen on validation (0.01 for DPSS, 0.001 for PCA). That *held-out* beam is the one whose test score is an independent measure. The chosen fit is then refitted on all samples and exported in the HFSS file format (*refit* and *exported*; their scores are in-sample, and the export changes the predictions by at most 0.25%). Outside 64.2–240.5 MHz the exported slices are the normalized HFSS beam.

4 Geometry

The positions are memo 002's geometry release `v0004` (`imgs/fits/v0004_marjum_geometry`: antenna in the 91 m era, transmitter as recommended for propagation); the mount and convention are memo 003's, with $\psi = 142.164°$. Two quantities are not known in advance: the azimuth offset added to the table's azimuth, and the transmitter's arm-0 angle α (the arm-0 field is $(-\sin\alpha, \cos\alpha, 0)$ in ENU, a compass bearing of $180° - \alpha$). A grid search with the HFSS beam finds them together (Fig. 2): offset $-16°$ and $\alpha = 96°$, normalized rms 0.122, against 0.330 at the best α with zero offset. Five small corrections are then fitted on top, in nested sets (polarization, azimuth and elevation zeros, transmitter east and north), and the set with the best validation score is kept (Table 1). All five are kept: $\delta\alpha = -0.15°$, $\delta\mathrm{az} = -0.04°$, $\delta\mathrm{el} = +0.93°$, and the transmitter moved by (+1.06, −1.06) m, which is the bound. The bound, $1.5/\sqrt{2}$ m per axis, is the package default (`tx_fit.DEFAULT_BOUNDS`), set from the historical `v0001` transmitter bracket of 1.5 m; `v0004` establishes no uncertainty to replace it. With the transmitter free the problem is poorly conditioned (condition number 64), and the position correction is a nuisance parameter, not a measurement.

Change from geometry v0001. The previous reference product, `empirical_raster_v0012`, is the same fit on the `v0001` antenna and the curated transmitter. In `v0004` the transmitter's direction from the antenna differs by 0.38° (4.23° to 4.61° from nadir), and by 0.37° once each product's fitted transmitter offsets are applied. The geometry fit absorbs this in the elevation zero, δel from +0.56° to +0.93°; the other corrections move by under 0.01°. The count-weighted test scores are the same to three decimals (HFSS 0.111, held-out DPSS 0.083; v0012 0.111 and 0.083), so the fit quality does not distinguish the two geometries. The elevation zero does depend on the geometry, which is one more reason it is not a measurement of the IMU offset.

Table 1: Nested geometry corrections, fitted with the HFSS beam on training samples and scored by the median per-tooth fractional rms on the validation and test stripes. The transmitter offsets are bounded at ±1.06 m.

free	$\delta\alpha$ [deg]	δaz [deg]	δel [deg]	δE [m]	δN [m]	cond.	val. / test median rms
none	+0.00	+0.00	+0.00	+0.00	+0.00	1	0.093 / 0.099
polarization	-0.21	+0.00	+0.00	+0.00	+0.00	1	0.093 / 0.099
az, el zero	+0.00	-0.04	+0.77	+0.00	+0.00	1	0.089 / 0.093
transmitter E, N	+0.00	+0.00	+0.00	+1.06	-1.06	2	0.089 / 0.094
polarization, az, el	-0.19	-0.02	+0.77	+0.00	+0.00	1	0.088 / 0.092
all five	-0.15	-0.04	+0.93	+1.06	-1.06	64	0.083 / 0.088

Agreement with memo 003. The offset is memo 003's −16° azimuth zero, measured there from the arm ratio alone, and the arm-0 bearing of 84° ($\alpha = 96°$) is memo 003's arm-0 bearing, along the transmitter axis photographed as rx1. Memo 003 shows that the arm ratio alone cannot tell the azimuth's handedness and that the photographed arm bearings favour the right-handed convention used here, with the transmitter's arm labels swapped. The beam pattern breaks the mirror symmetry slightly, so the fit can test it directly: rerunning the geometry stage with the table's azimuth negated (Table 2) gives the mirrored solution, arm 0 at 171° near rx6, and fits worse at every stage (coarse rms 0.129 against 0.122; test 0.093 against 0.088). The evidence is weak,

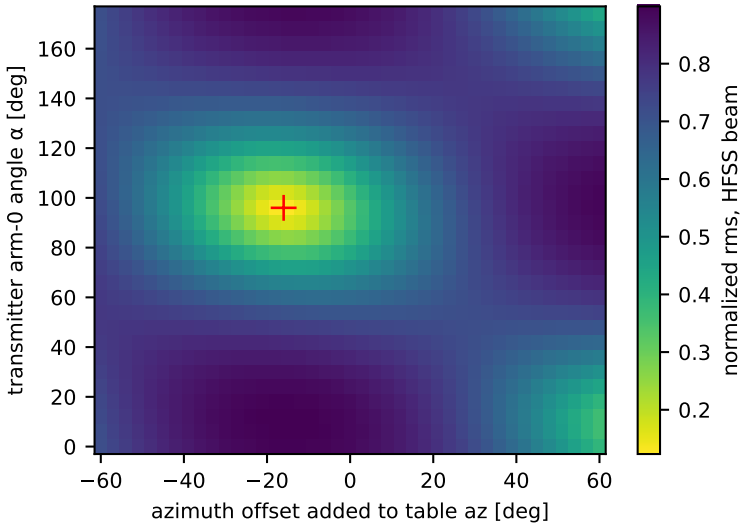


Figure 2: The HFSS beam's fit to the training samples over the azimuth offset and the transmitter arm-0 angle α; the cross marks the minimum.

but it points the same way as memo 003. Beam fits before v0010 used a different convention and fitted slightly better than v0010; the comparison above, on this product's data and code, does not reproduce that preference.

Table 2: The geometry stage with each azimuth handedness: the coarse scan's best azimuth zero, the bearing of transmitter arm 0 and its rms, then the median per-tooth fractional rms on the validation and test stripes with no corrections and with all five.

azimuth handedness	az zero [deg]	arm 0 [deg]	coarse rms	no corrections val. / test	all corrections val. / test
right-hand	-16	84	0.122	0.093 / 0.099	0.083 / 0.088
left-hand	+16	171	0.129	0.100 / 0.105	0.089 / 0.093

5 Results

Figure 3 shows the data and both models for two teeth. Both follow the large modulation as the boresight passes the transmitter once per revolution; the differences are in the shoulders and the sidelobes.

Raster maps. The direct comparison is the raster itself (Fig. 4): each tooth's background-subtracted power at every good sample, placed at its azimuth and elevation in the product's fitted geometry (table azimuth plus the $-16°$ offset and δaz, IMU elevation plus δel), beside the HFSS prediction at the same samples with that tooth's gain fitted on the training stripes, and their difference. Appendix B shows all 101 candidate teeth, 50.8–246.1 MHz, one by one, including the 37 the selection dropped (Fig. 1); for those the HFSS gain is fitted the same way and nothing else is. The product holds only the selected teeth, so the excluded ones are rebuilt from the raster with the product's own generator (`fit_beam.load_data`: same samples, background and masks), which reproduces the selected teeth to <0.001% rms of each tooth.

Scores. On the test stripes the held-out DPSS beam reaches 0.083 (90th percentile 0.133, worst 0.173 at 132.8 MHz) against HFSS's 0.111 (0.182, 0.321 at 230.5 MHz) (Table 3). PCA does the same (0.083). The empirical beam is better on 59 of 64 teeth (Fig. 5). The worst empirical tooth, 132.8 MHz, is one of the weak arm-0 teeth, whose background matters most.

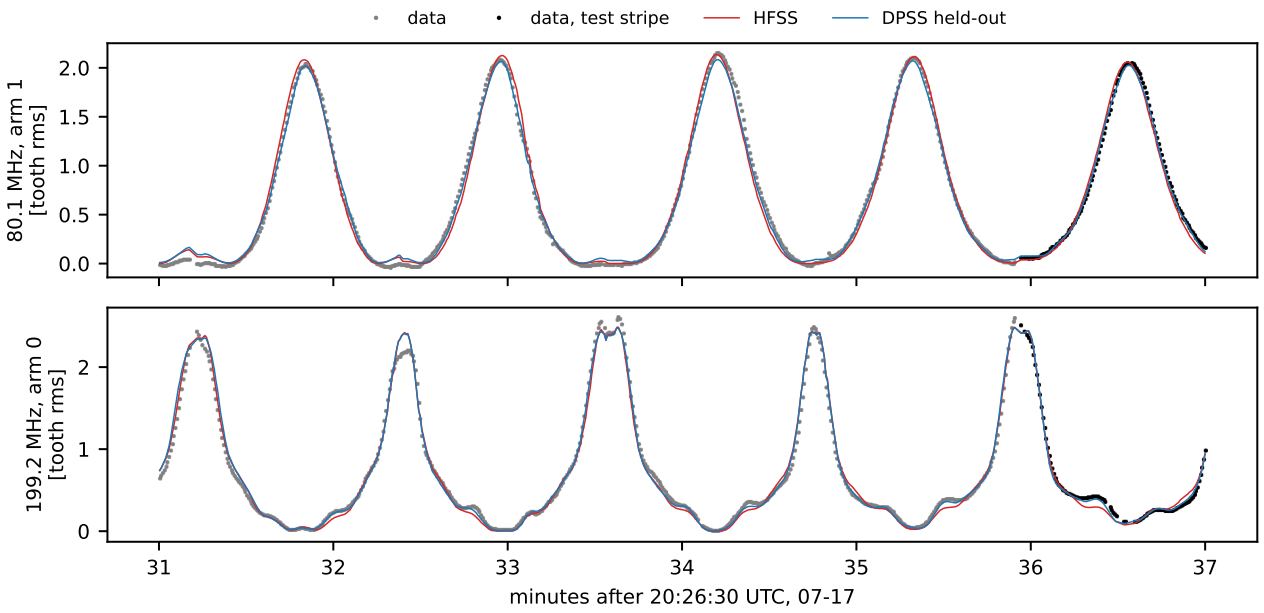


Figure 3: Two teeth over six minutes of the raster, in units of each tooth's rms: data (grey; black on a test stripe), the HFSS beam with its fitted gain (red), and the held-out DPSS beam (blue). Each peak is one elevation revolution passing the transmitter.

Table 3: Per-tooth fractional rms $\|d-m\|/\|d\|$ over the 64 teeth: the mean weighted by good-sample count, the 90th percentile and the worst tooth. Only the held-out fits' test column is independent of every fitted choice; HFSS has one gain per tooth fitted on the training stripes.

model	train	validation	test
HFSS	0.107 / 0.173 / 0.285	0.102 / 0.167 / 0.275	0.111 / 0.182 / 0.321
DPSS held-out	0.080 / 0.119 / 0.162	0.088 / 0.125 / 0.241	0.083 / 0.133 / 0.173
PCA held-out	0.079 / 0.112 / 0.164	0.087 / 0.123 / 0.242	0.083 / 0.124 / 0.175
DPSS refit	0.081 / 0.121 / 0.164	0.084 / 0.119 / 0.235	0.080 / 0.130 / 0.162
DPSS exported	0.081 / 0.121 / 0.164	0.084 / 0.120 / 0.235	0.080 / 0.130 / 0.162

The residual is not noise. The white-noise part of each tooth's residual, estimated from differences of successive samples, is 0.014 of the tooth's rms on average (worst 0.065, the weak 132.8 MHz arm-0 tooth). The test residual is ∼6× larger, so it is structure that changes slowly along the raster: beam features the model does not capture, background errors, or geometry. Binned by the transmitter's angle from the boresight (Fig. 6), the fractional residual is smallest near the boresight and near the back of the antenna, where most of the tooth power is, and largest at 60–140°, where the power is one to two orders of magnitude lower. The empirical beam improves on HFSS at every angle.

The beam. Figure 7 compares the exported DPSS beam with HFSS at three frequencies. Weighting each sampled pixel by its number of visits and by the beam's power there, the two differ by 0.2–0.5 dB rms, and at least 99% of that weight is within 1 dB; single low-power pixels differ by up to 9 dB. The raster samples 44% of the HEALPix pixels; elsewhere the beam is set by the HFSS prior and the smoothness of the basis, and its differences from HFSS there are not measurements.

6 Caveats

- **Phase and circular polarization** are not constrained: the data are powers. The coefficients are regularized toward HFSS, which fixes what the data cannot.

Table 4: Test-stripe fractional rms by band (mean over teeth weighted by sample count), and the white-noise floor.

band [MHz]	teeth	HFSS	DPSS held-out	PCA held-out	noise floor
50–87	6	0.089	0.063	0.071	0.011
87–145	11	0.097	0.085	0.087	0.017
145–190	23	0.076	0.064	0.065	0.011
190–235	22	0.154	0.102	0.098	0.016
235–245	2	0.196	0.133	0.134	0.019

- **The test is a held-out stripe of azimuth, not a fresh experiment.** The stripes share the session, the background model and the geometry with the training data.
- **Background**: the transmitter-off control test of the background (about 1% bias, 2–3% scatter) was made with the earlier 15-mode background and has not been rerun for the 21-mode background used since v0012; it matters most for the weak arm-0 teeth. The raster flags were built with an earlier beam (v0006) as reference.
- **Geometry**: the transmitter offsets are at a bound inherited from `v0001`, `v0004` has no established uncertainty, and the fitted elevation zero (+0.93°) moved by about the change in the transmitter's direction; it does not match the IMU offset implied by the LIDAR (memo 003).
- **One tooth regressed** from v0011 to v0012: 68.4 MHz test score 0.057 to 0.098, unexplained (v0012 README).
- The $\lambda = 10^{-4}$ fits reached the 3000-iteration limit; the chosen fits converged.

7 Open questions

- What the remaining structure is: a higher $\ell_{\max}$ or a richer spectral basis where the residual is largest (60–140°), against background and geometry errors.
- The transmitter's arm labels (memo 003): the beam fit and the arm ratio both put arm 0 along the axis photographed as rx1.
- A transmitter-off control for the 21-mode background.

Revision history

2026-10-08 Moved to beam product v0013, the same fit on geometry release `v0004`; adds the comparison with v0012's `v0001` geometry.

2026-10-02 First version, on beam product v0012. Adds count-weighted scores with worst cases, the white-noise floor, the residual by angle, and the handedness check of the geometry stage.

A Reproducing this memo

The beam product is regenerated, from the eigsep checkout root, by `python data-analysis/scripts/marjum-2026-07/fit_beam.py -output marjum-2026-07/derived/beam/empirical_raster_vNNNN`, with its defaults (teeth `tooth_selection_v0002.json`, flags `raster_flags_v0001.npz`, DPSS and PCA bases, 3000 iterations, background concentration 10^{-6}); the commits it ran at are below. This memo's figures, tables and numbers come from `python make_products.py` in this directory (set `EIGSEP_CAMPAIGN_ROOT` to the campaign directory that holds `derived/beam/` when running from a worktree), then `pdflatex memo-005.tex` twice; set `EIGSEP_DATA_ANALYSIS` to use a data-analysis checkout other than the one beside `eigsep_data`. The script checks the SHA-256 of every gitignored input.

input	SHA-256 (first 16)
derived/beam/empirical_raster_v0013/provenance.json	37b9f6b2432200c3
derived/beam/empirical_raster_v0013/metrics.csv	0b7d48cce7a620e4
derived/beam/empirical_raster_v0013/geometry_comparison.csv	7dc4789060318810
derived/beam/empirical_raster_v0013/az_offset_alpha_scan.csv	432cd664cc67caab
derived/beam/empirical_raster_v0013/spatial_order.csv	5c00257ee7f503a9
derived/beam/empirical_raster_v0013/dpss/diagnostics.npz	c829b29129c0ac6c
derived/beam/empirical_raster_v0013/dpss/empirical_beam.npz	4c4de71693e14b97
derived/beam/empirical_raster_v0013/dpss/basis_selection.csv	57d1a3941d9777e1
derived/beam/empirical_raster_v0013/pca/diagnostics.npz	9b428a4f568065ab
derived/beam/empirical_raster_v0013/pca/basis_selection.csv	f254c86fe09d1c2a
derived/beam/empirical_raster_v0012/provenance.json	d1221e804109cde4
derived/beam/empirical_raster_v0012/dpss/diagnostics.npz	fcfc12f9da087629
derived/beam/tooth_selection_v0002.csv	acd4e893bc6c1899
derived/beam/tooth_selection_v0002.json	08e4d5e3e48ee9e5
eigsep_data/hfss_beam_maps/bowtie_beam.npz	c4046f5e9332cd57
marjum-2026-07/imgs/fits/v0004_marjum_geometry/shared.json	8fcf5360cbfa14a0
marjum-2026-07/imgs/fits/v0001_marjum_geometry/shared.json	234c1d5ab7142318
beam product generated by data-analysis/scripts/marjum-2026-07/fit_beam.py	
data-analysis	5c6a072-dirty
eigsep_data	e941c51
eigsep_base	840c10b
eigsep	821f4fd
eigsep (memo) commit	01ce21c
eigsep_data commit (HFSS file)	e941c51
data-analysis commit (raster rebuild, fit_beam.py)	37092de-dirty
generated	2026-10-08 21:38 UTC

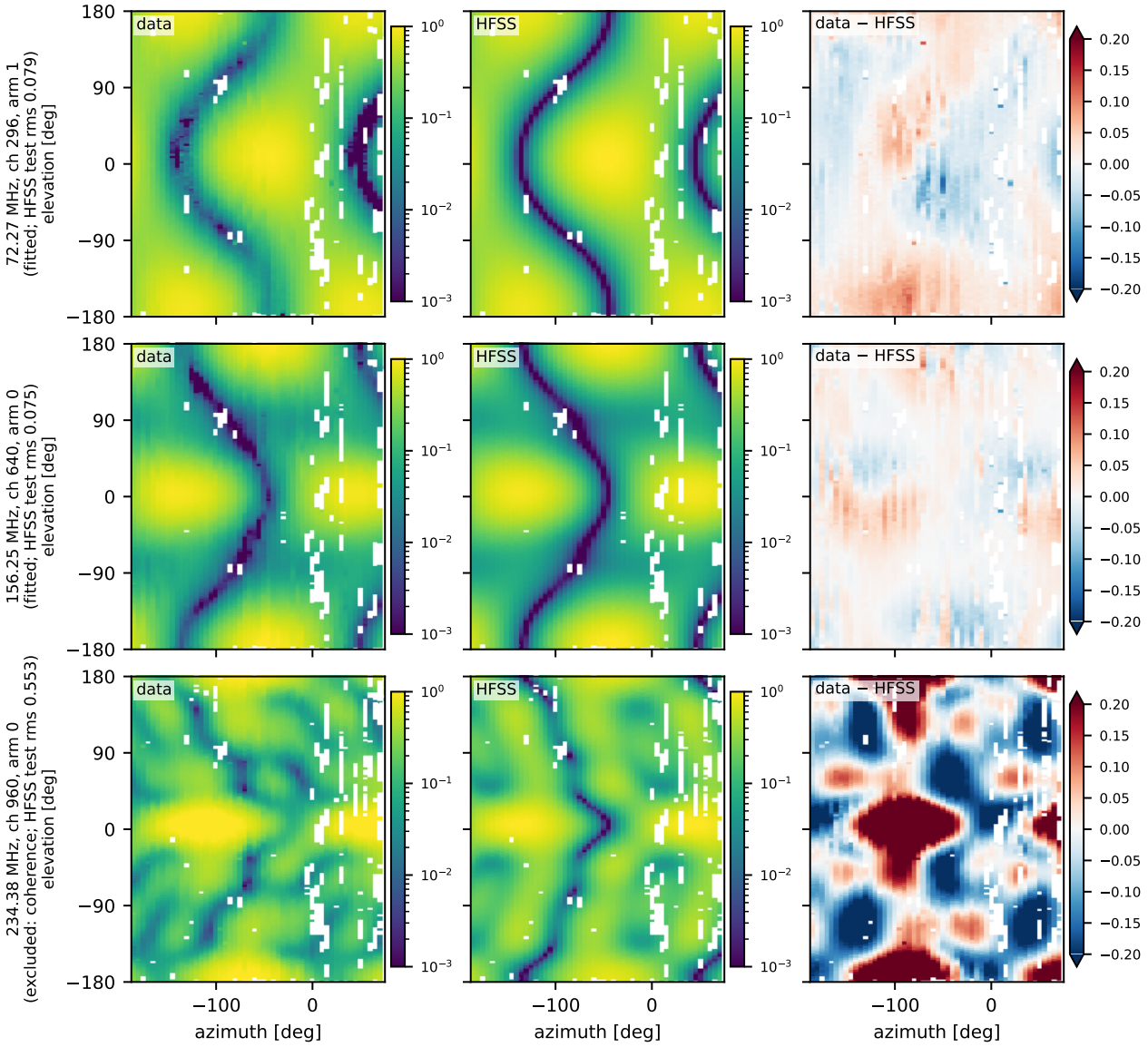


Figure 4: Raster maps of three teeth on azimuth (from the highline, fitted geometry) and elevation (0 at zenith, $\pm 180°$ at nadir): data (tooth minus background), the HFSS model with one fitted gain, and data minus model. Every panel is divided by the HFSS model's peak over the raster (its fitted gain times its peak coupling), so all teeth share fixed scales: data and model on a log scale from 10^{-3} to 1 (lower values are drawn at the floor), the residual from -0.2 to $+0.2$ of the peak. A good fit leaves a pale residual panel; a poor one saturates. Only good samples are drawn. The labels give each tooth's HFSS test-stripe fractional rms; 234.38 MHz was excluded from the fit by the coherence test.

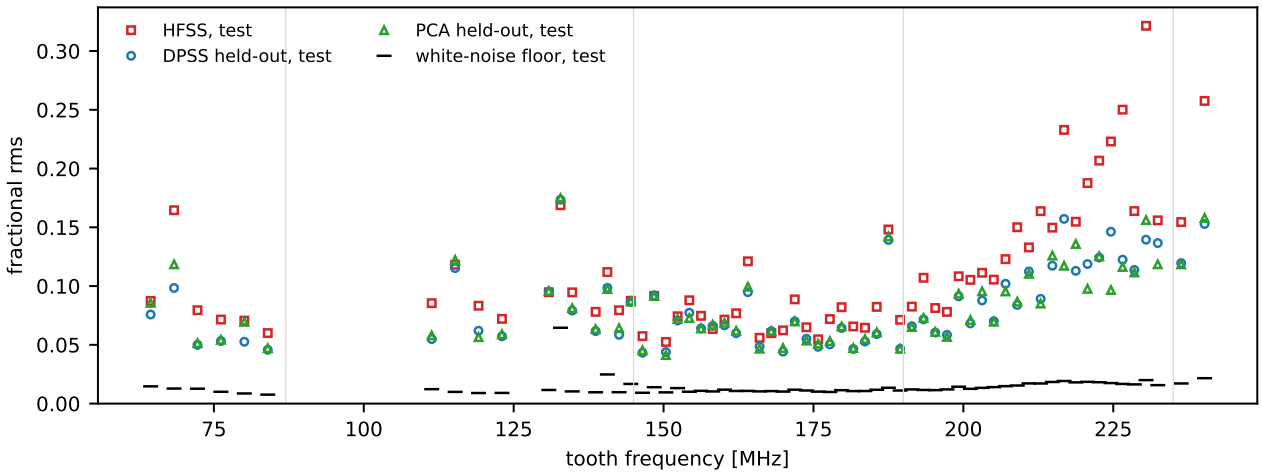


Figure 5: Per-tooth fractional rms on the test stripes for HFSS and the two held-out empirical beams, with the white-noise floor of each tooth's residual. Thin lines mark the band edges of Table 4.

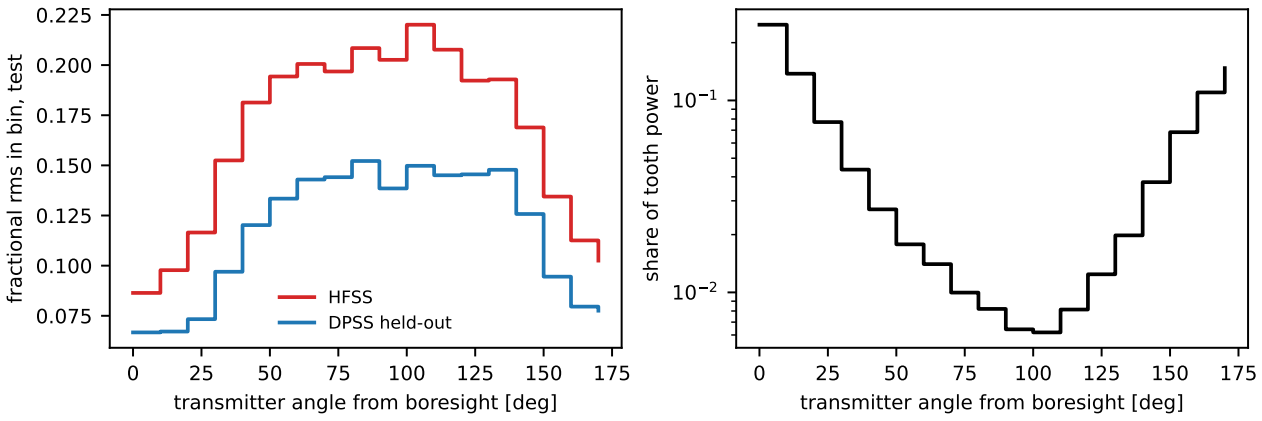


Figure 6: Left: fractional rms on the test stripes, pooled over teeth (each tooth scaled to unit mean power), in 10° bins of the transmitter's angle from the boresight. Right: the share of tooth power in each bin. The antenna has no ground plane, so it receives about as much from behind as from in front.

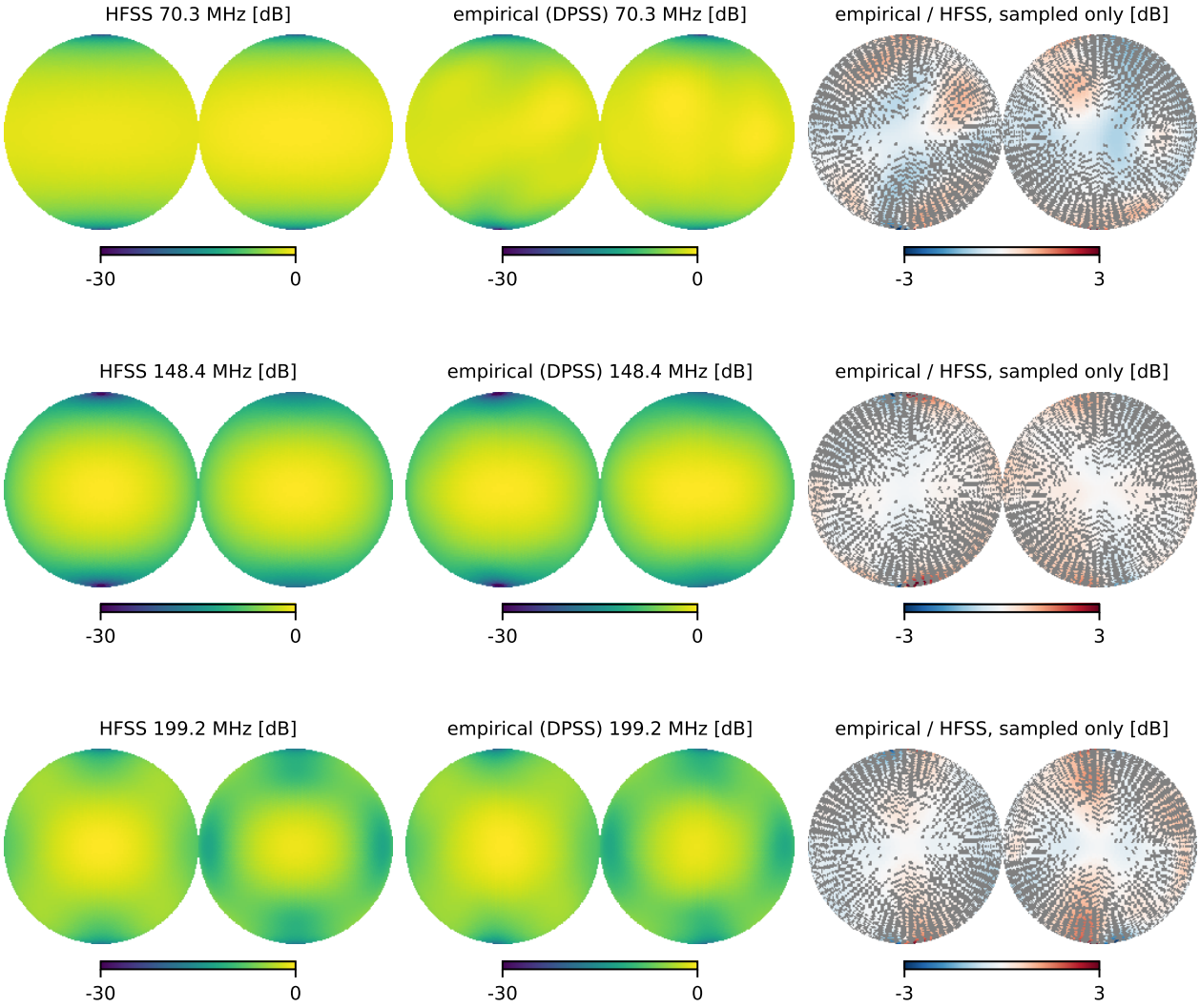


Figure 7: Total power $|\mathbf{E}|^2$ of HFSS (left) and the exported DPSS beam (centre), each peak-normalized, and their ratio on the sampled pixels only (right), at three frequencies. Each panel shows the forward hemisphere (boresight at the centre of the left disc) and the backward hemisphere (right disc).

B Raster maps of every candidate tooth

All 101 candidate teeth in frequency order, 26 pages of four, with the panels, geometry and colour scales of Fig. 4. Each row's label says whether the tooth was fitted or why it was excluded, and gives its HFSS test-stripe fractional rms. Teeth in the FM band (87–108.5 MHz) carry broadcast signals as well as the transmitter, and the background there is fitted without the FM channels. Where the fitted gain is zero (gains are constrained non-negative) there is no HFSS peak to divide by; those rows are divided by the data's 99.5th percentile instead (marked "gain 0"), their HFSS panel sits at the floor, and the test rms is 1. Where the gain is merely small, the scaled data run far off the top of the scale.

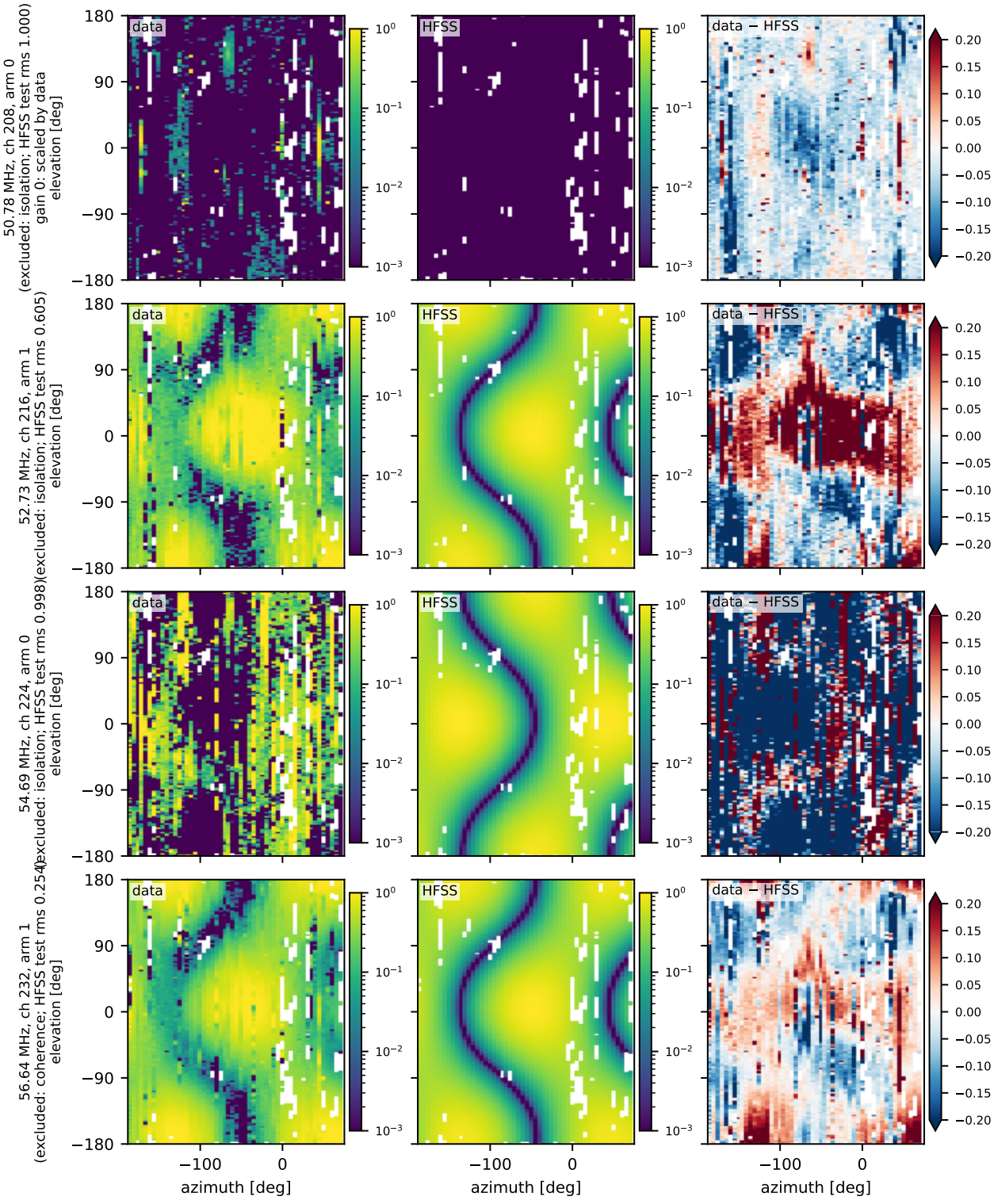


Figure 8: Raster maps, 50.78–56.64 MHz (page 1 of 26); panels as in Fig. 4.

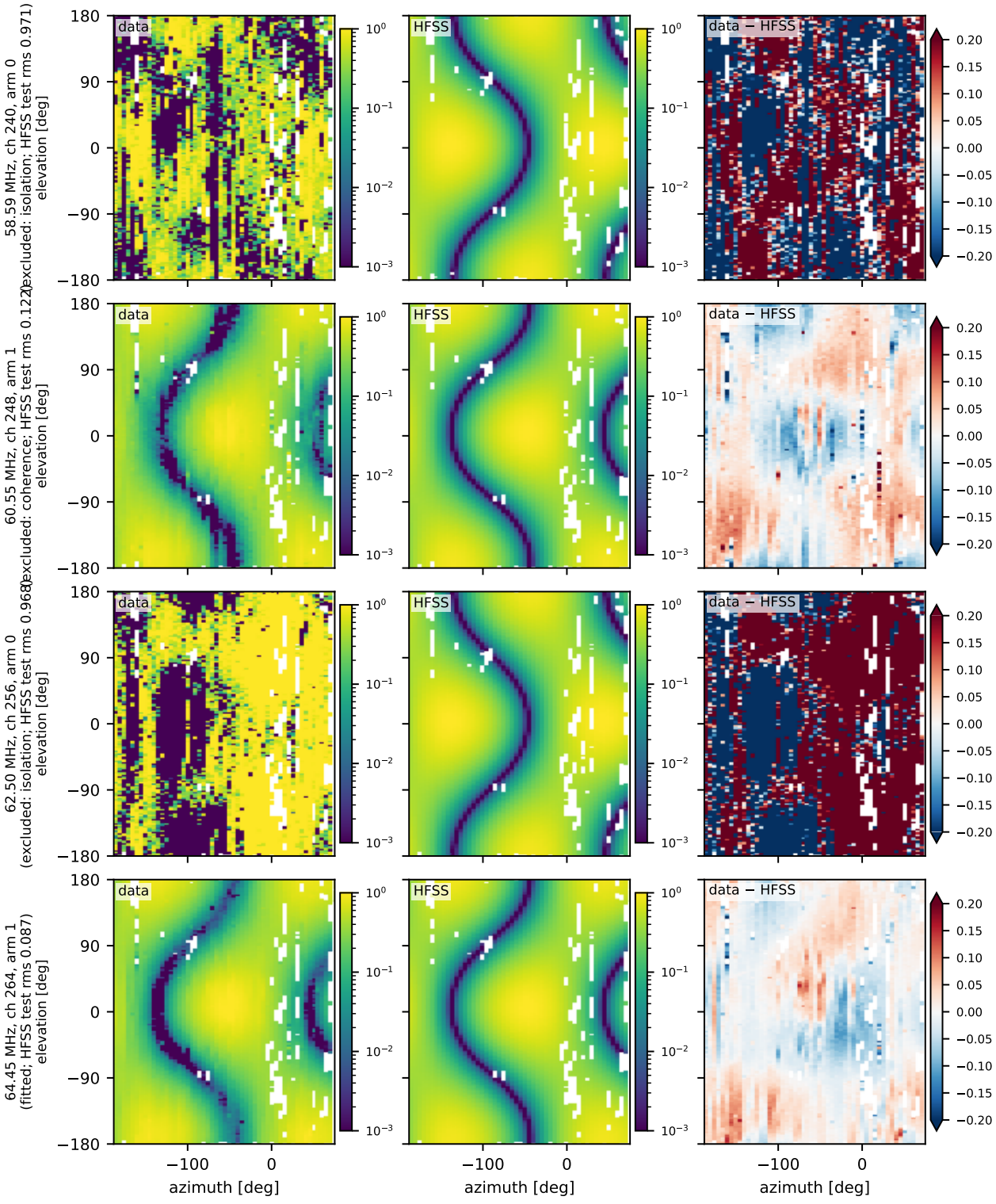


Figure 9: Raster maps, 58.59–64.45 MHz (page 2 of 26); panels as in Fig. 4.

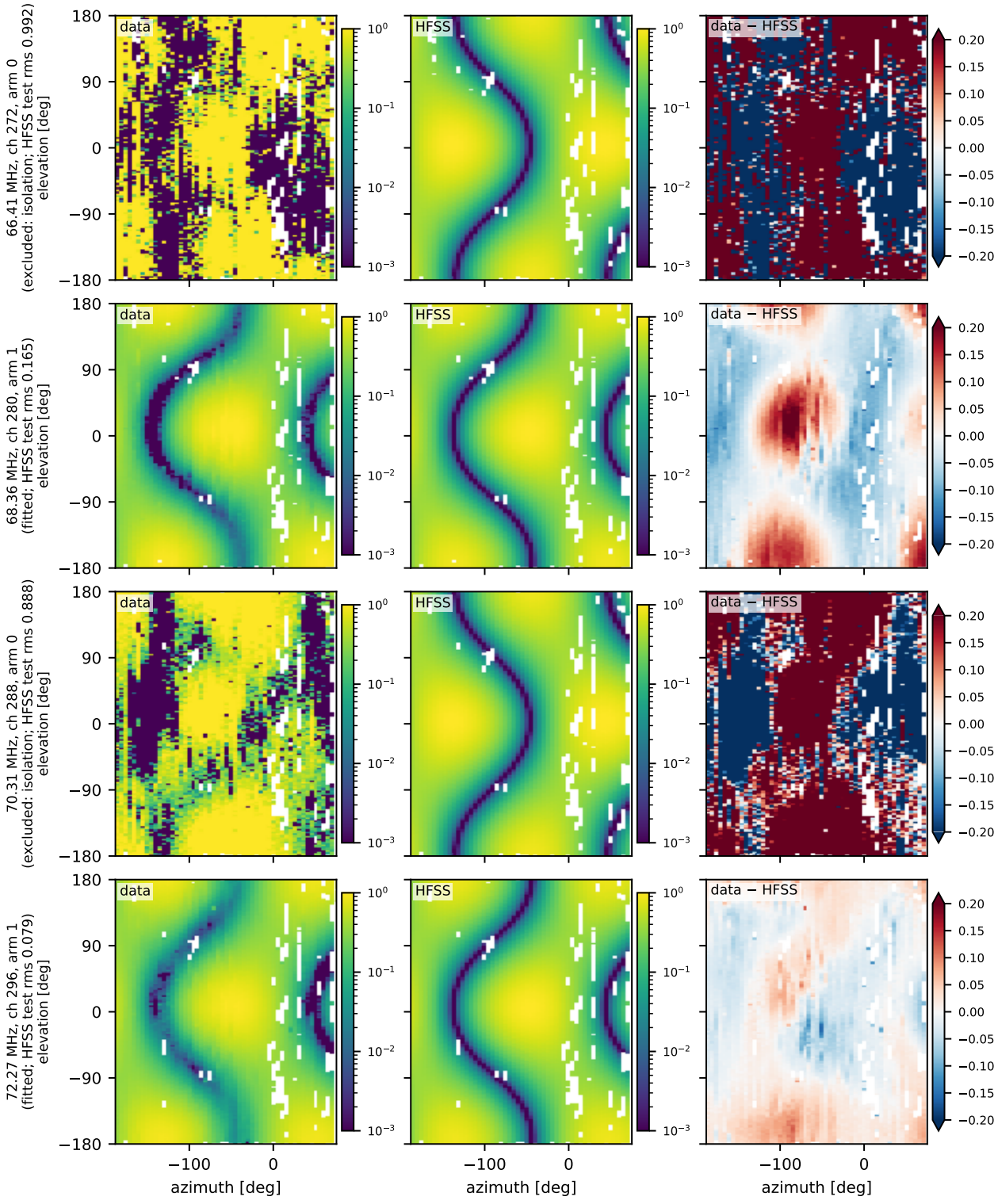


Figure 10: Raster maps, 66.41–72.27 MHz (page 3 of 26); panels as in Fig. 4.

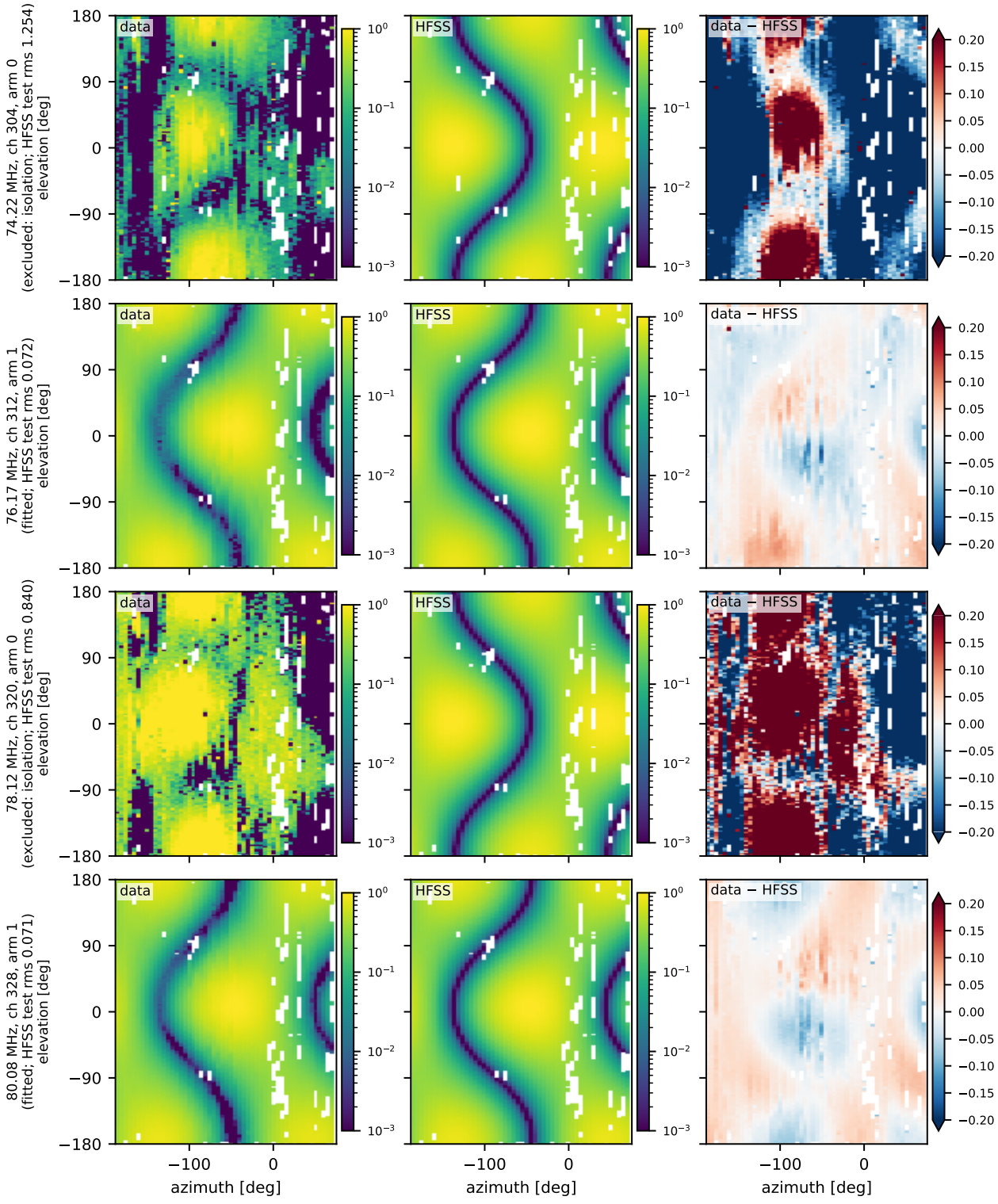


Figure 11: Raster maps, 74.22–80.08 MHz (page 4 of 26); panels as in Fig. 4.

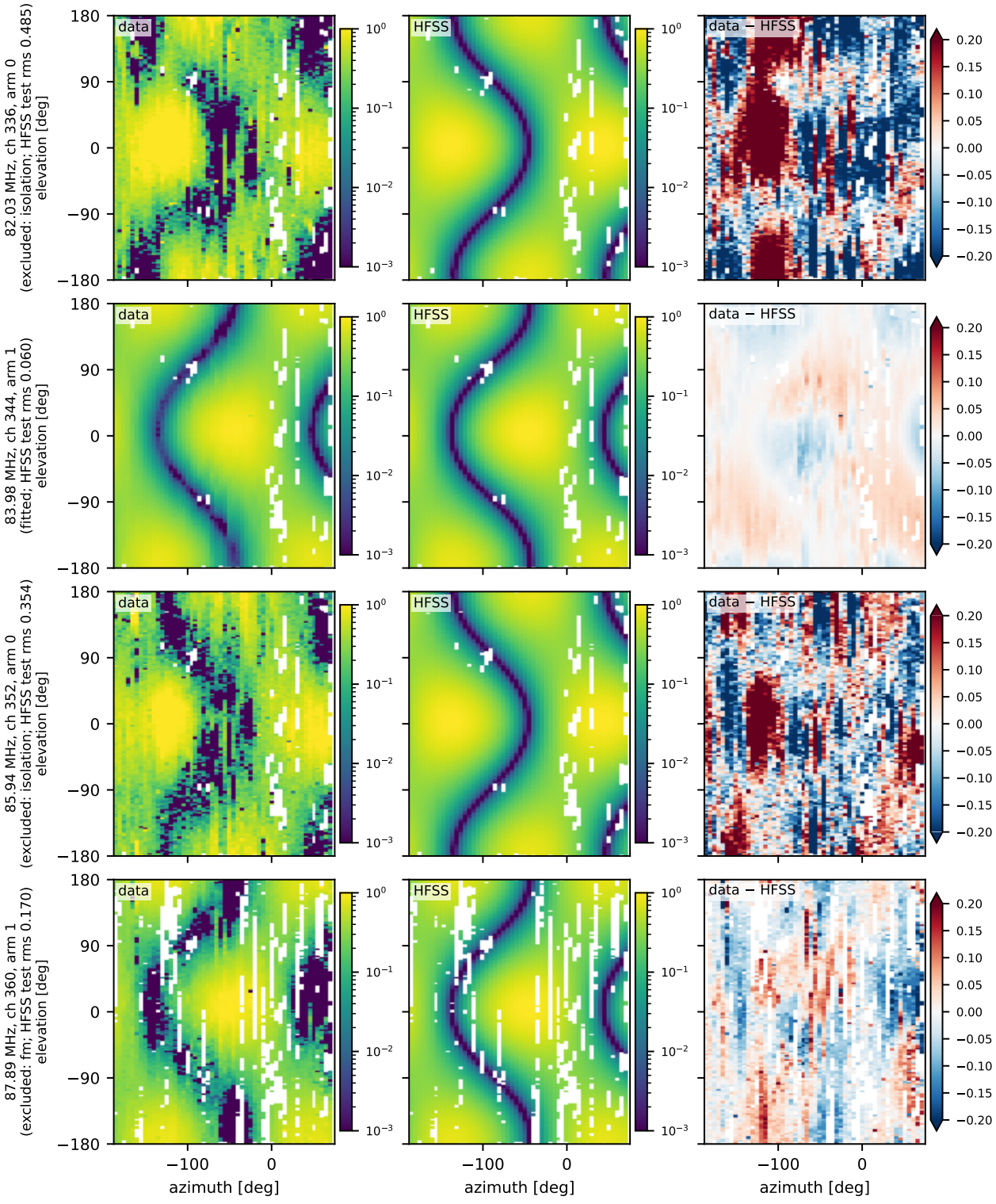


Figure 12: Raster maps, 82.03–87.89 MHz (page 5 of 26); panels as in Fig. 4.

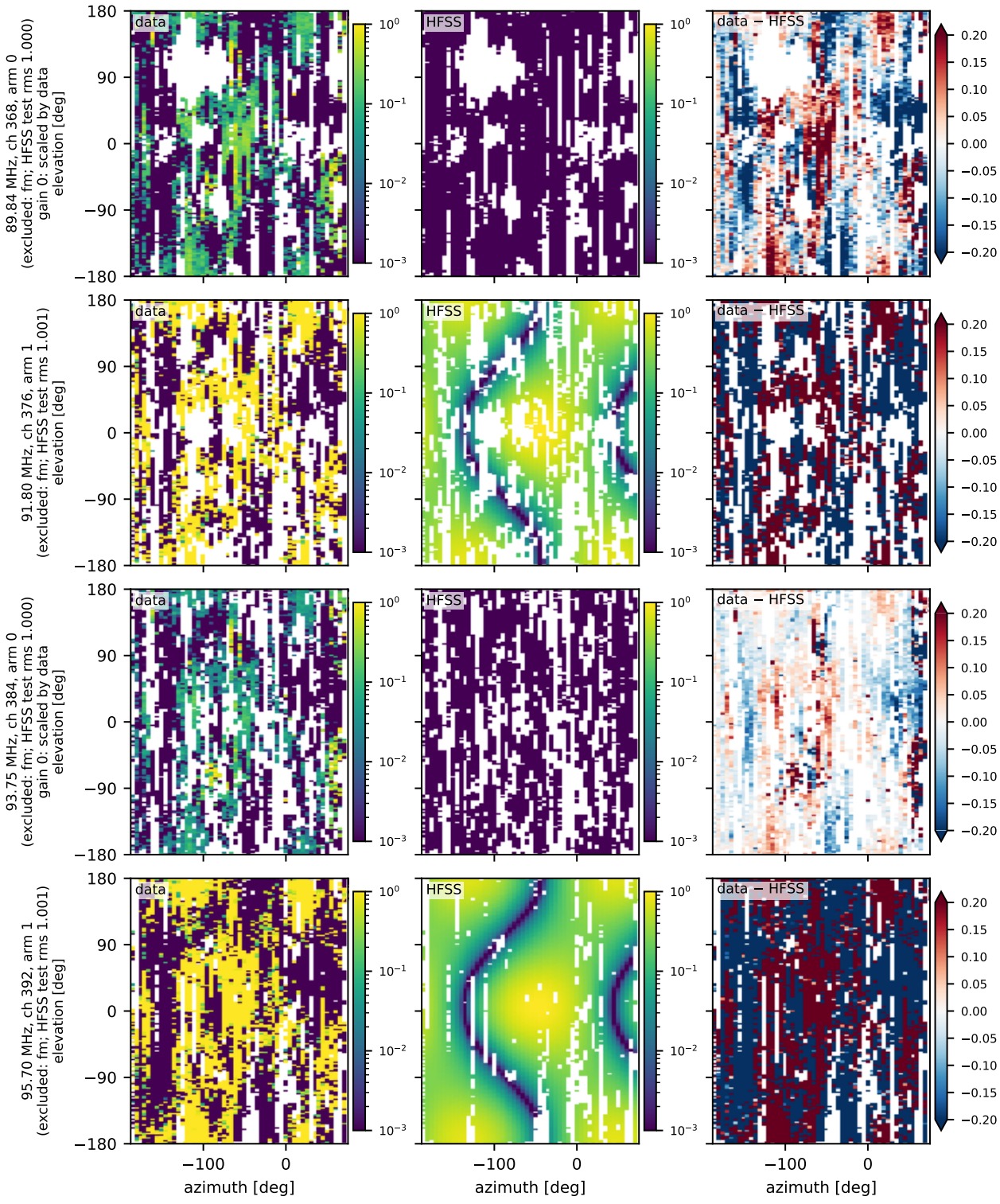


Figure 13: Raster maps, 89.84–95.70 MHz (page 6 of 26); panels as in Fig. 4.

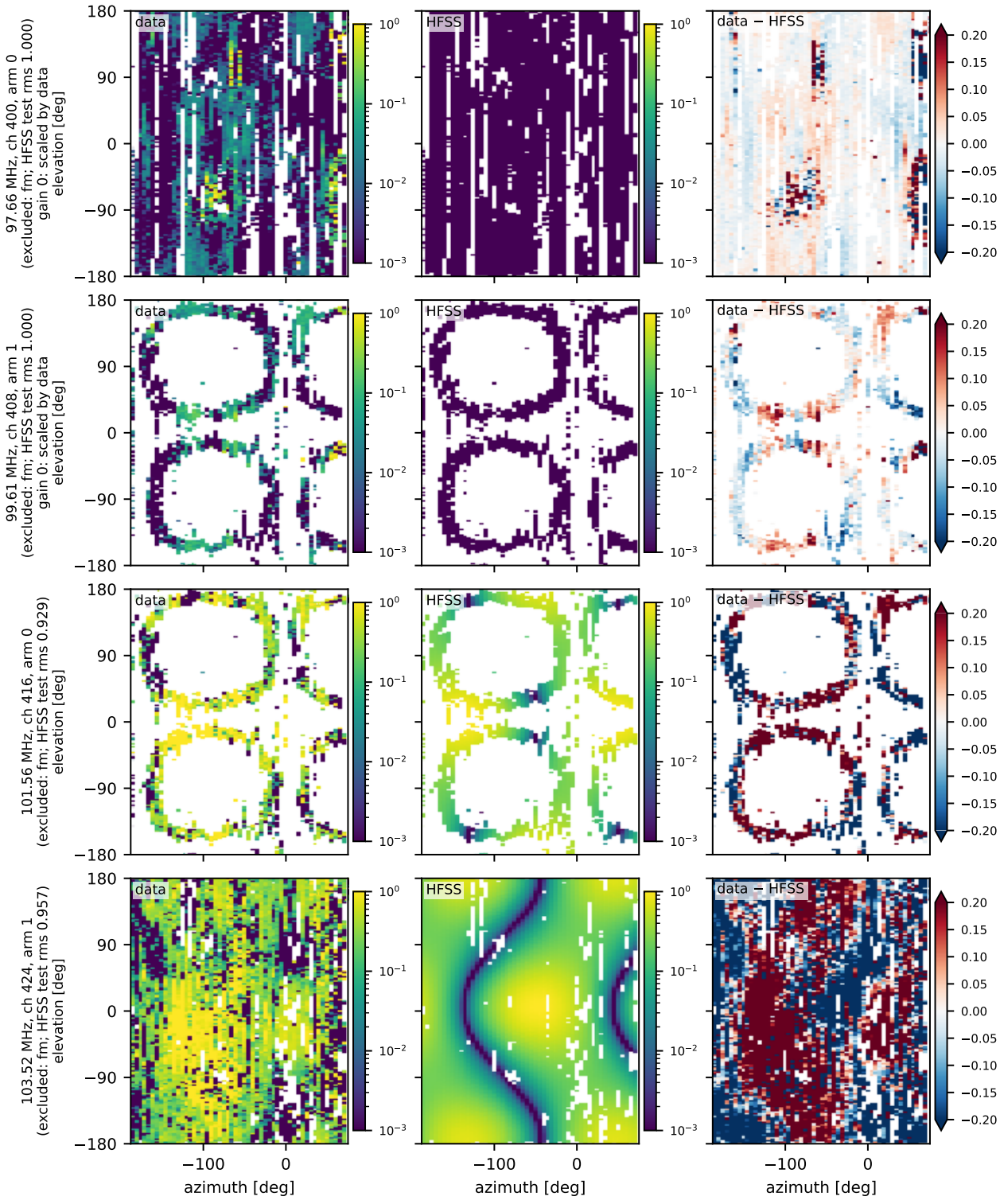


Figure 14: Raster maps, 97.66–103.52 MHz (page 7 of 26); panels as in Fig. 4.

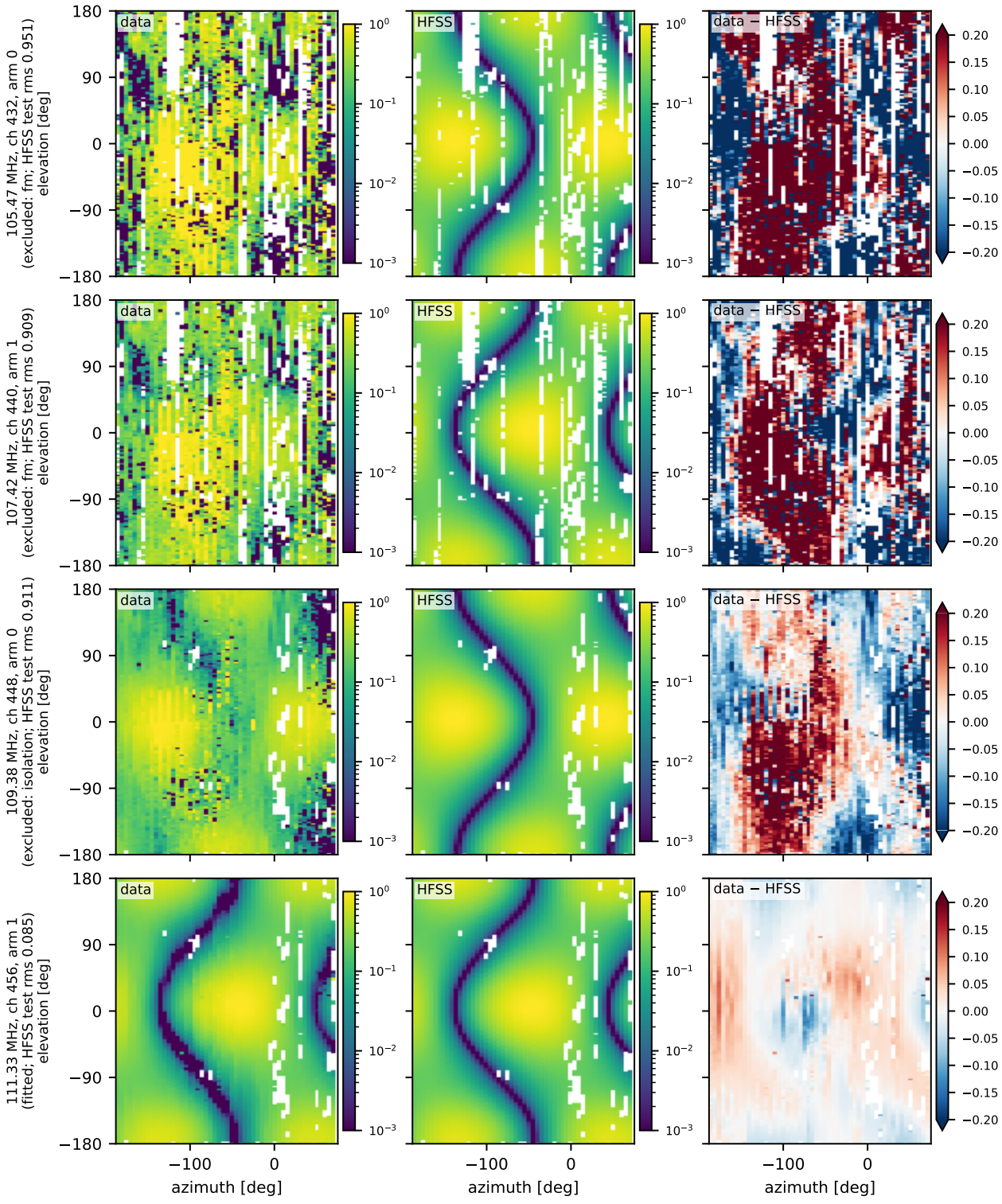


Figure 15: Raster maps, 105.47–111.33 MHz (page 8 of 26); panels as in Fig. 4.

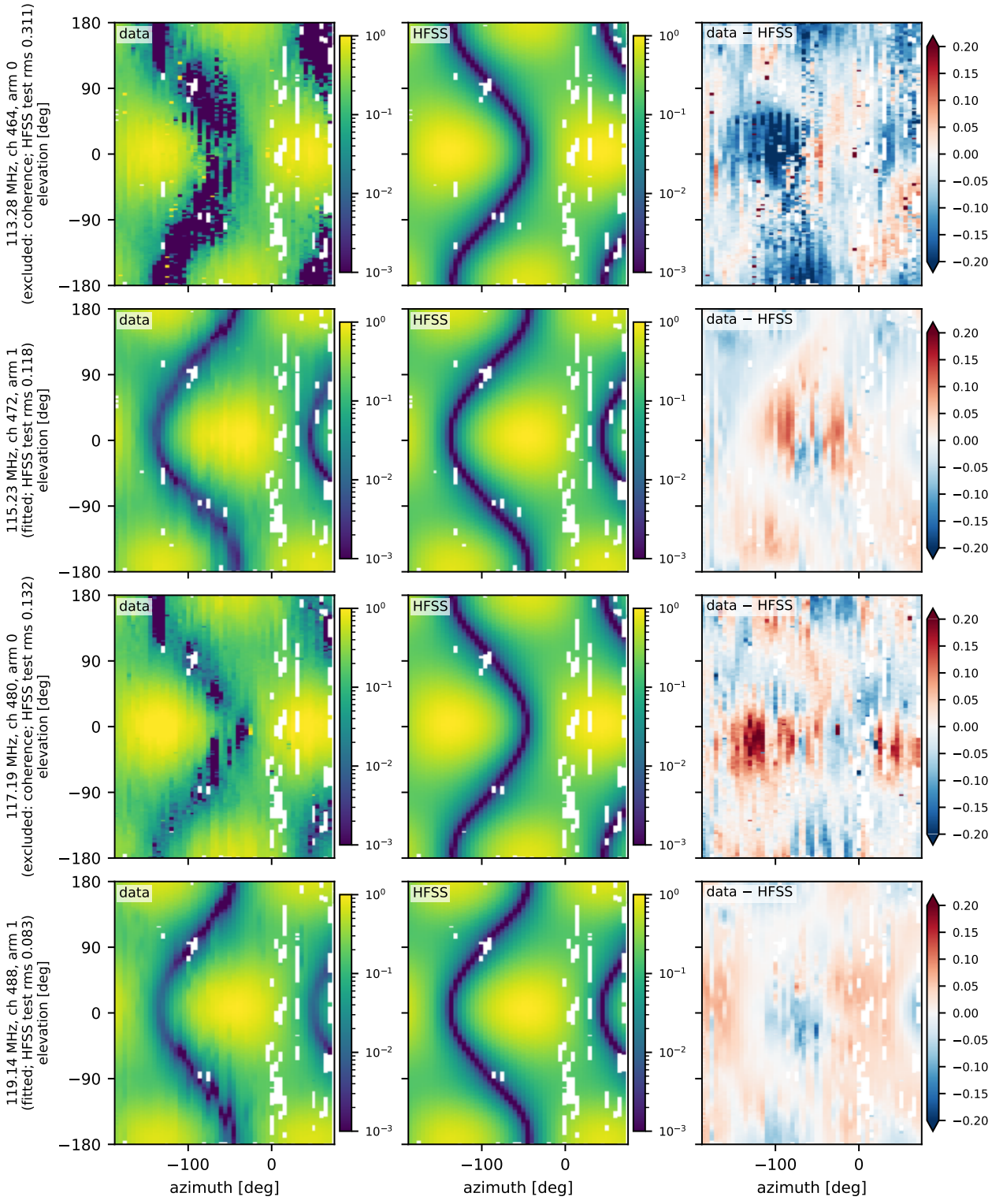


Figure 16: Raster maps, 113.28–119.14 MHz (page 9 of 26); panels as in Fig. 4.

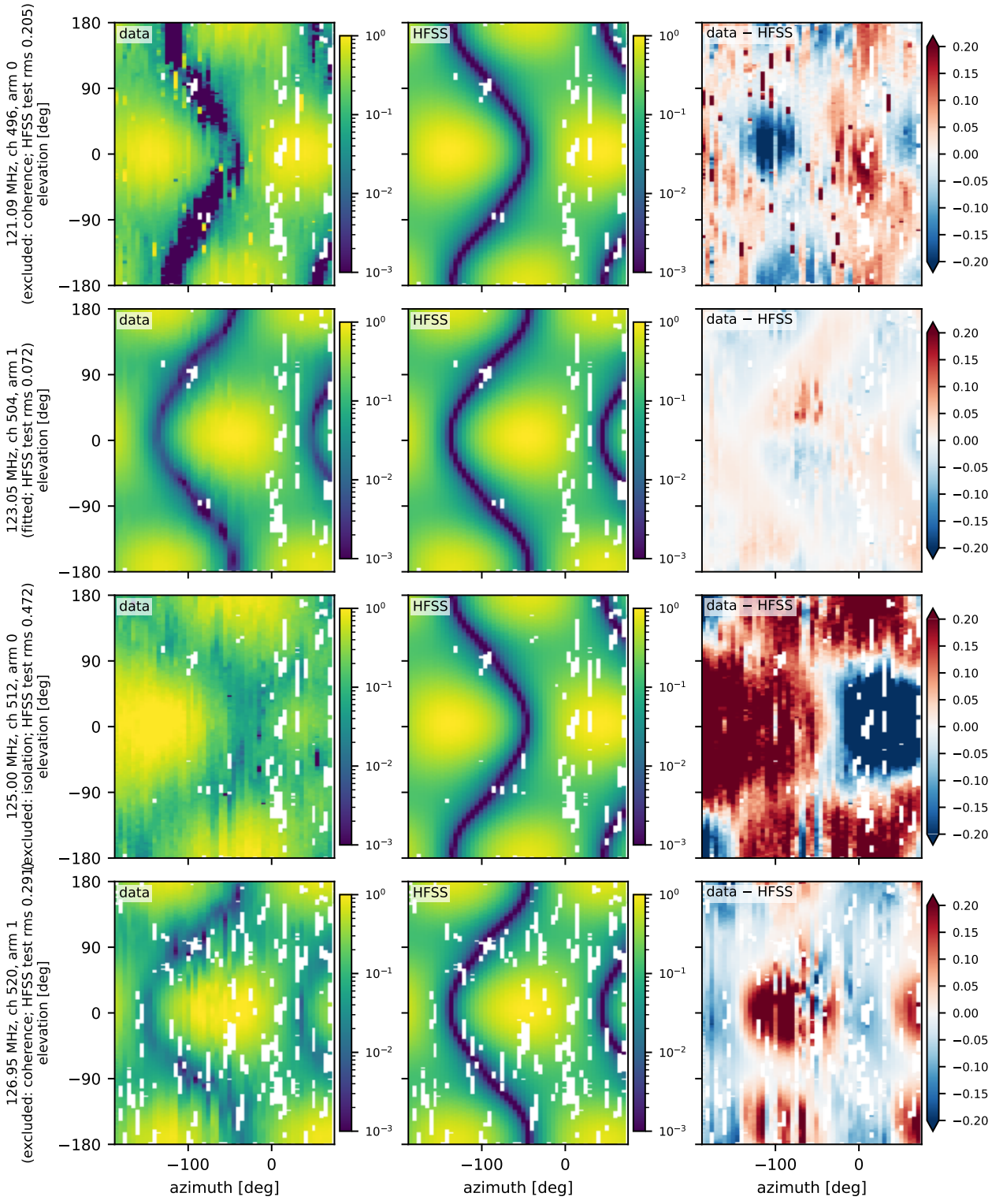


Figure 17: Raster maps, 121.09–126.95 MHz (page 10 of 26); panels as in Fig. 4.

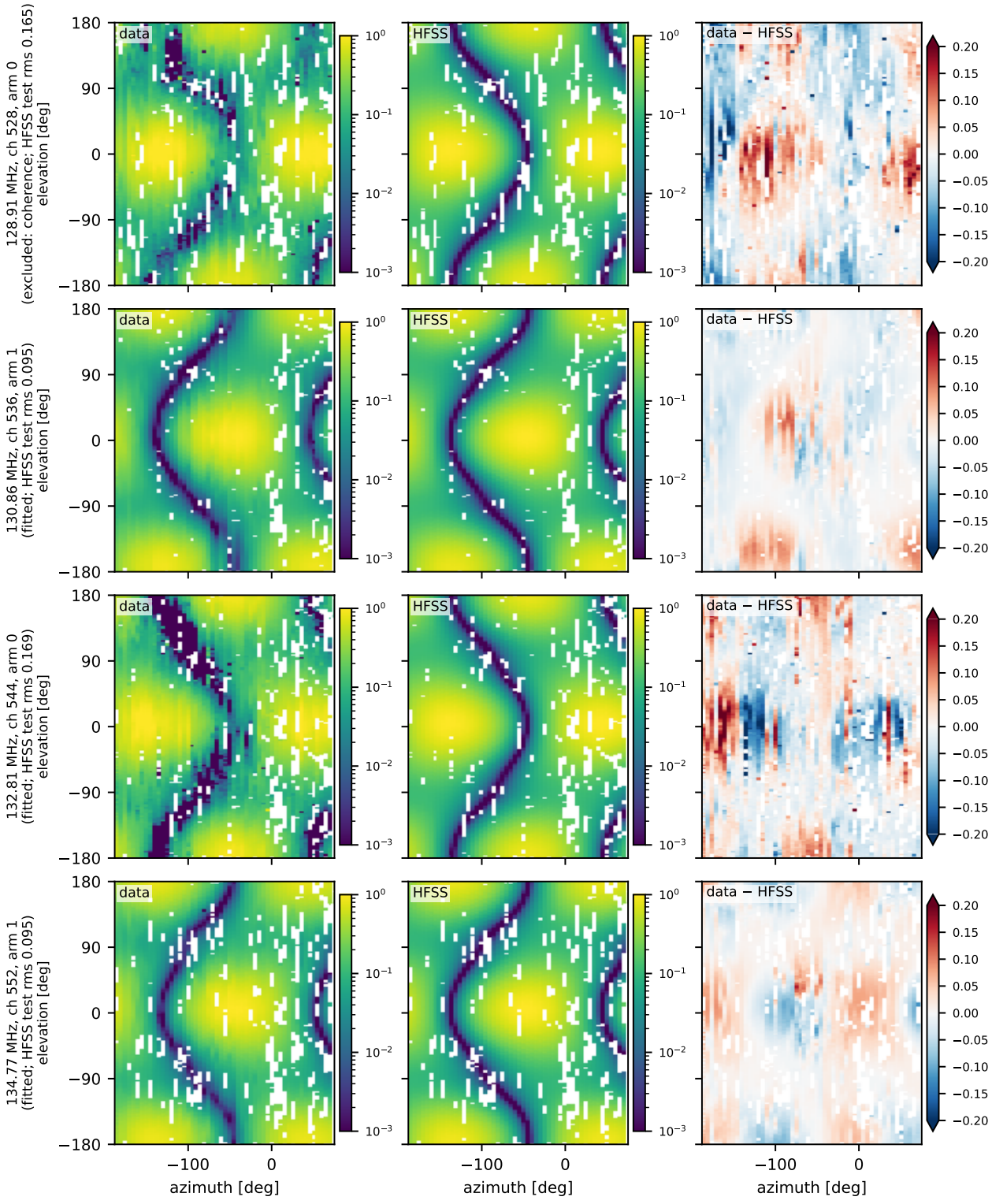


Figure 18: Raster maps, 128.91–134.77 MHz (page 11 of 26); panels as in Fig. 4.

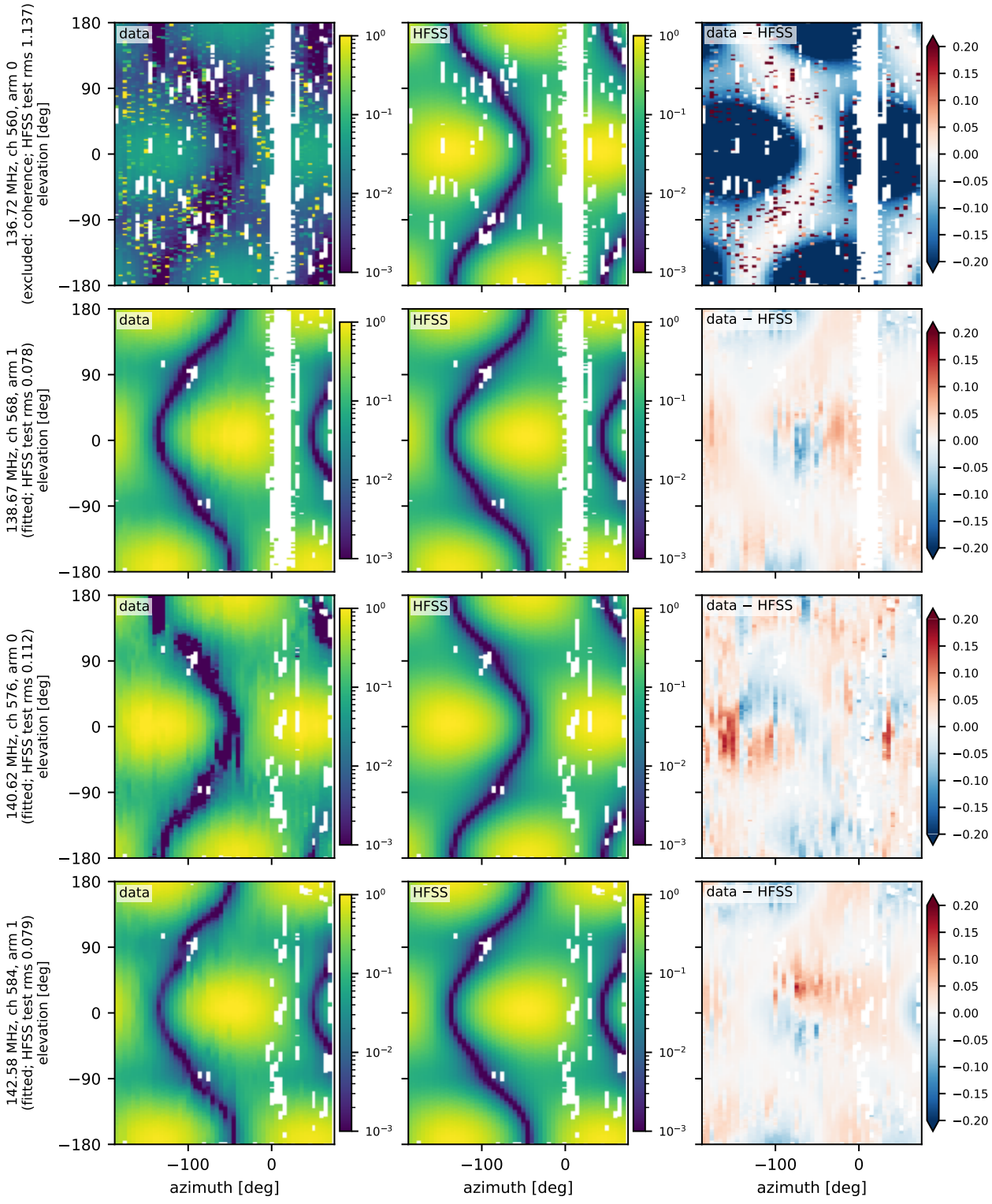


Figure 19: Raster maps, 136.72–142.58 MHz (page 12 of 26); panels as in Fig. 4.

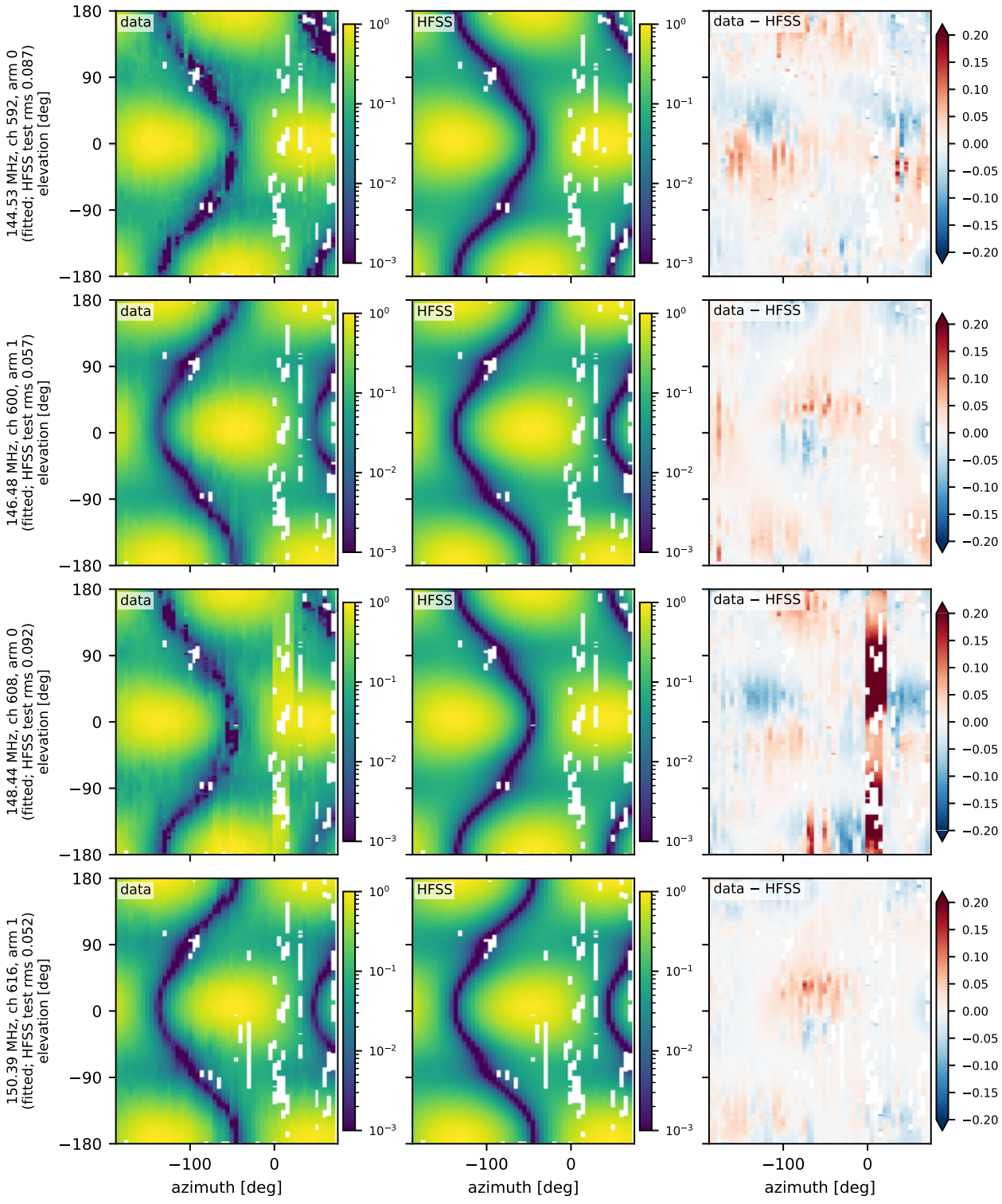


Figure 20: Raster maps, 144.53–150.39 MHz (page 13 of 26); panels as in Fig. 4.

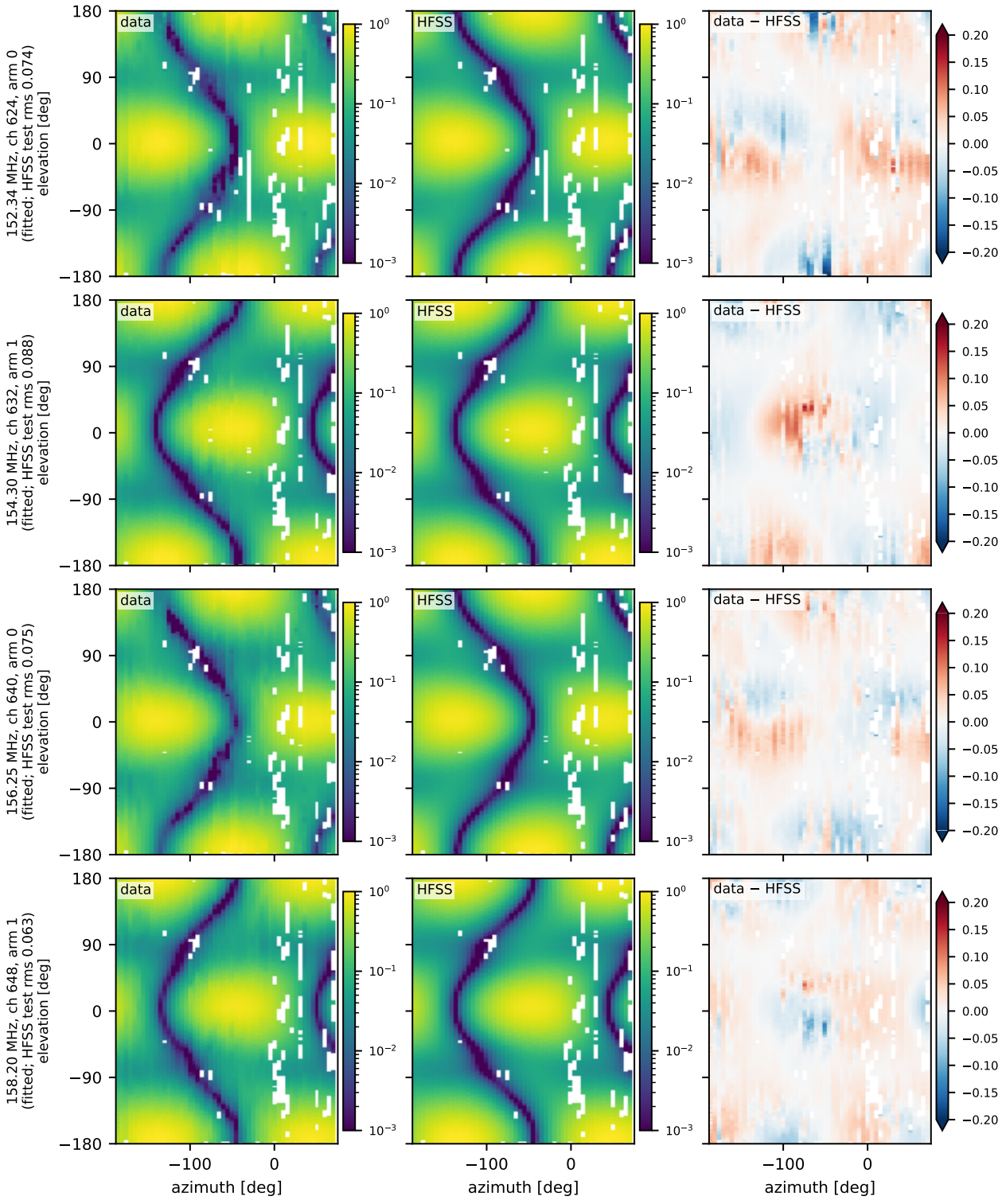


Figure 21: Raster maps, 152.34–158.20 MHz (page 14 of 26); panels as in Fig. 4.

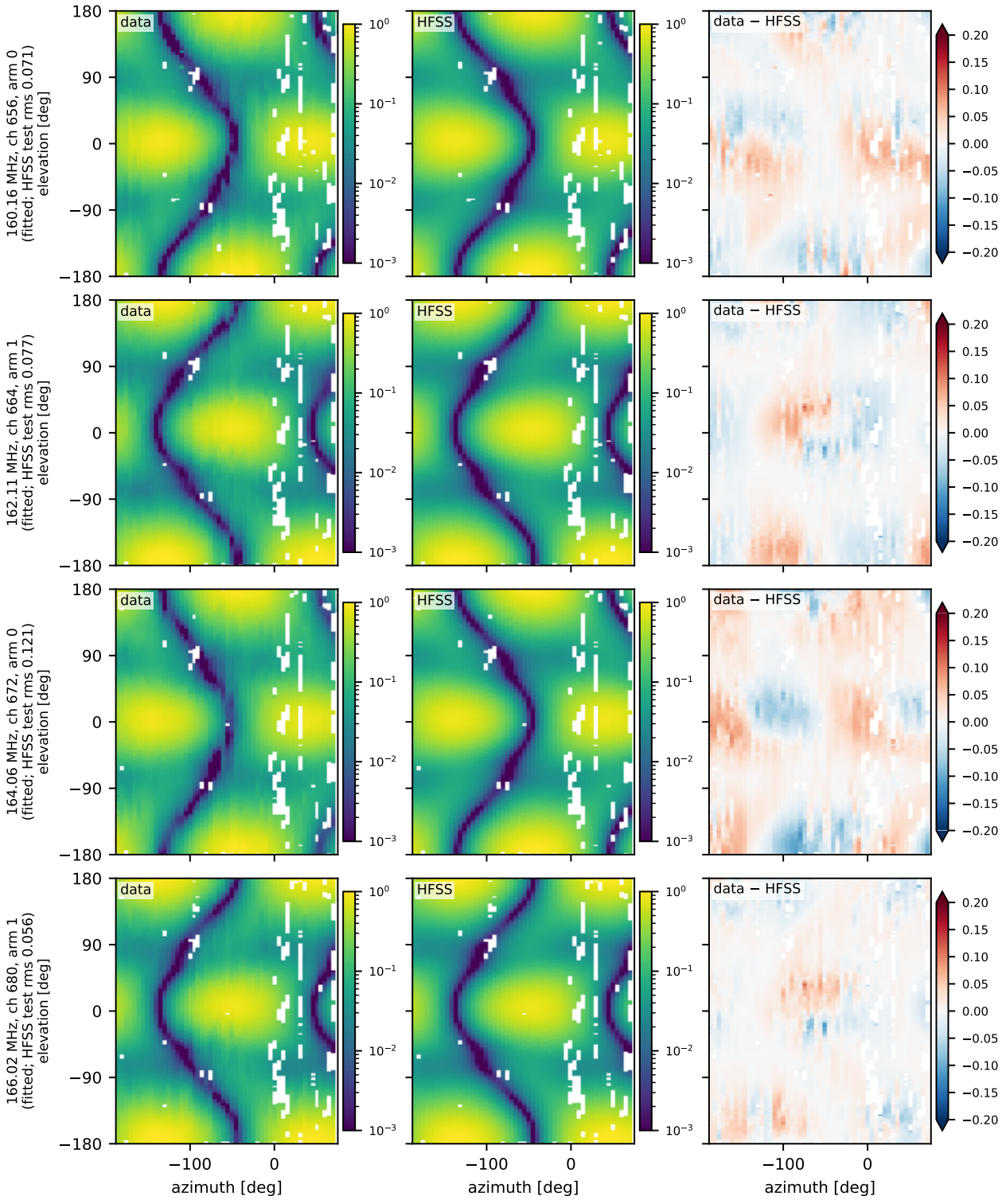


Figure 22: Raster maps, 160.16–166.02 MHz (page 15 of 26); panels as in Fig. 4.

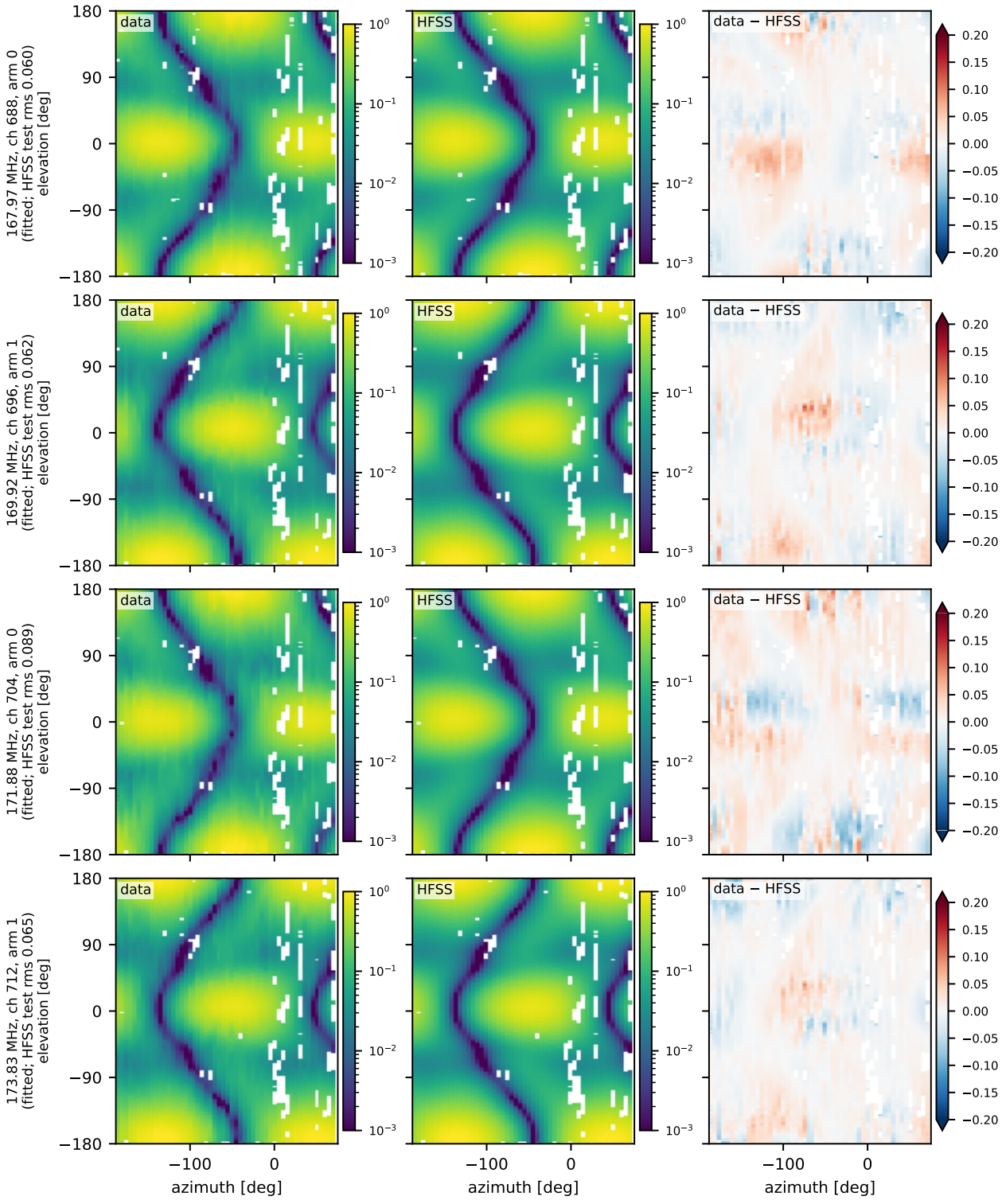


Figure 23: Raster maps, 167.97–173.83 MHz (page 16 of 26); panels as in Fig. 4.

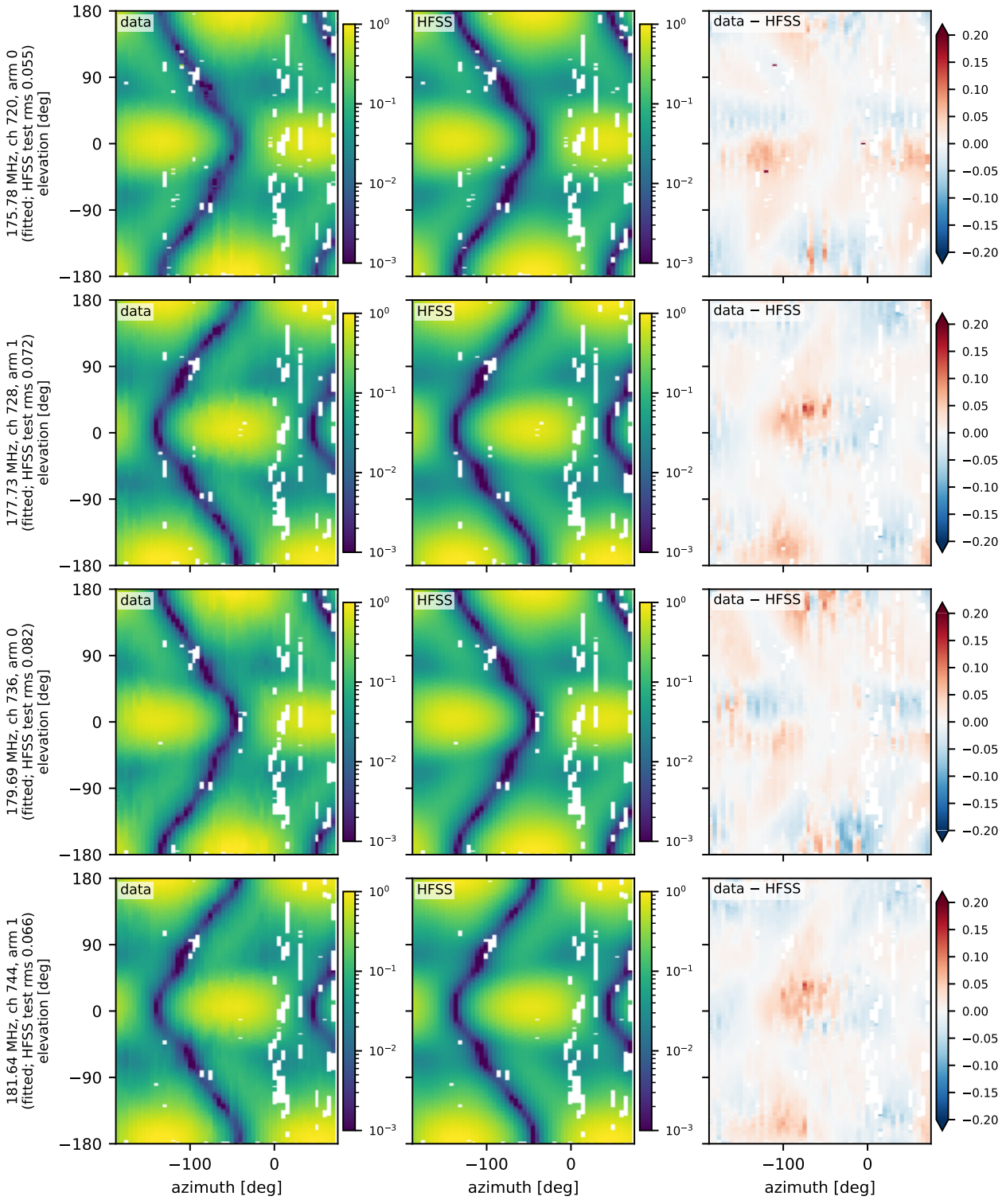


Figure 24: Raster maps, 175.78–181.64 MHz (page 17 of 26); panels as in Fig. 4.

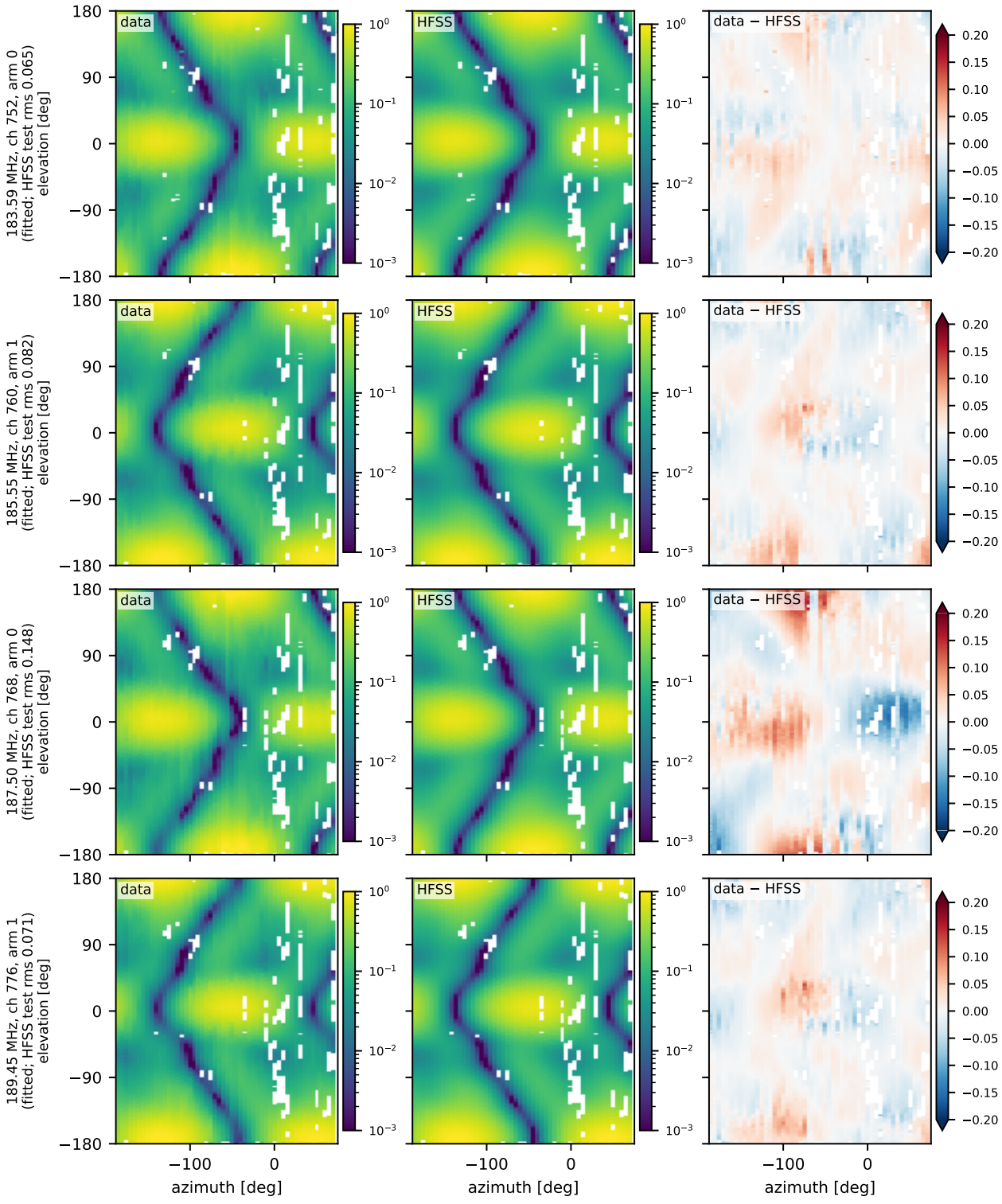


Figure 25: Raster maps, 183.59–189.45 MHz (page 18 of 26); panels as in Fig. 4.

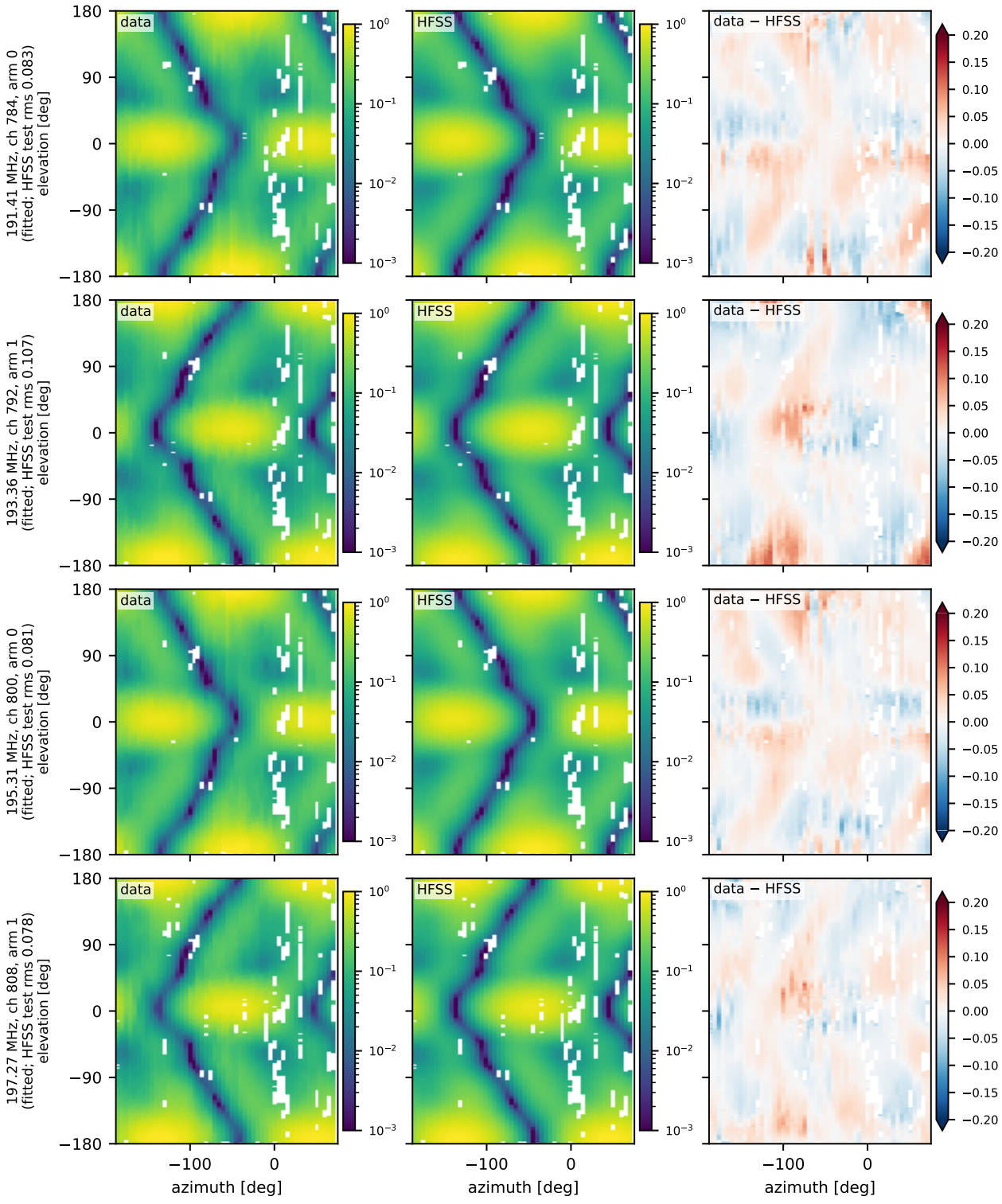


Figure 26: Raster maps, 191.41–197.27 MHz (page 19 of 26); panels as in Fig. 4.

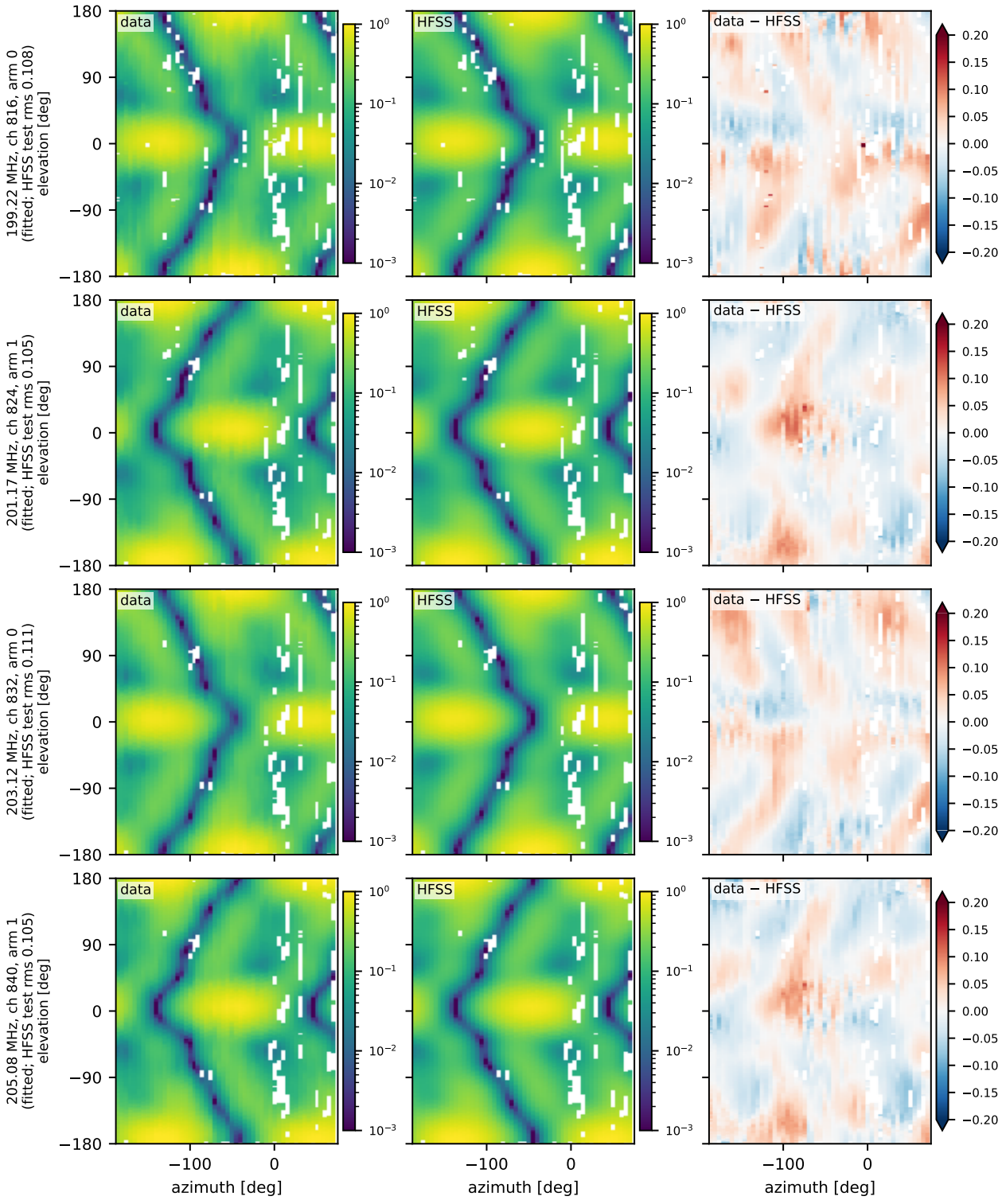


Figure 27: Raster maps, 199.22–205.08 MHz (page 20 of 26); panels as in Fig. 4.

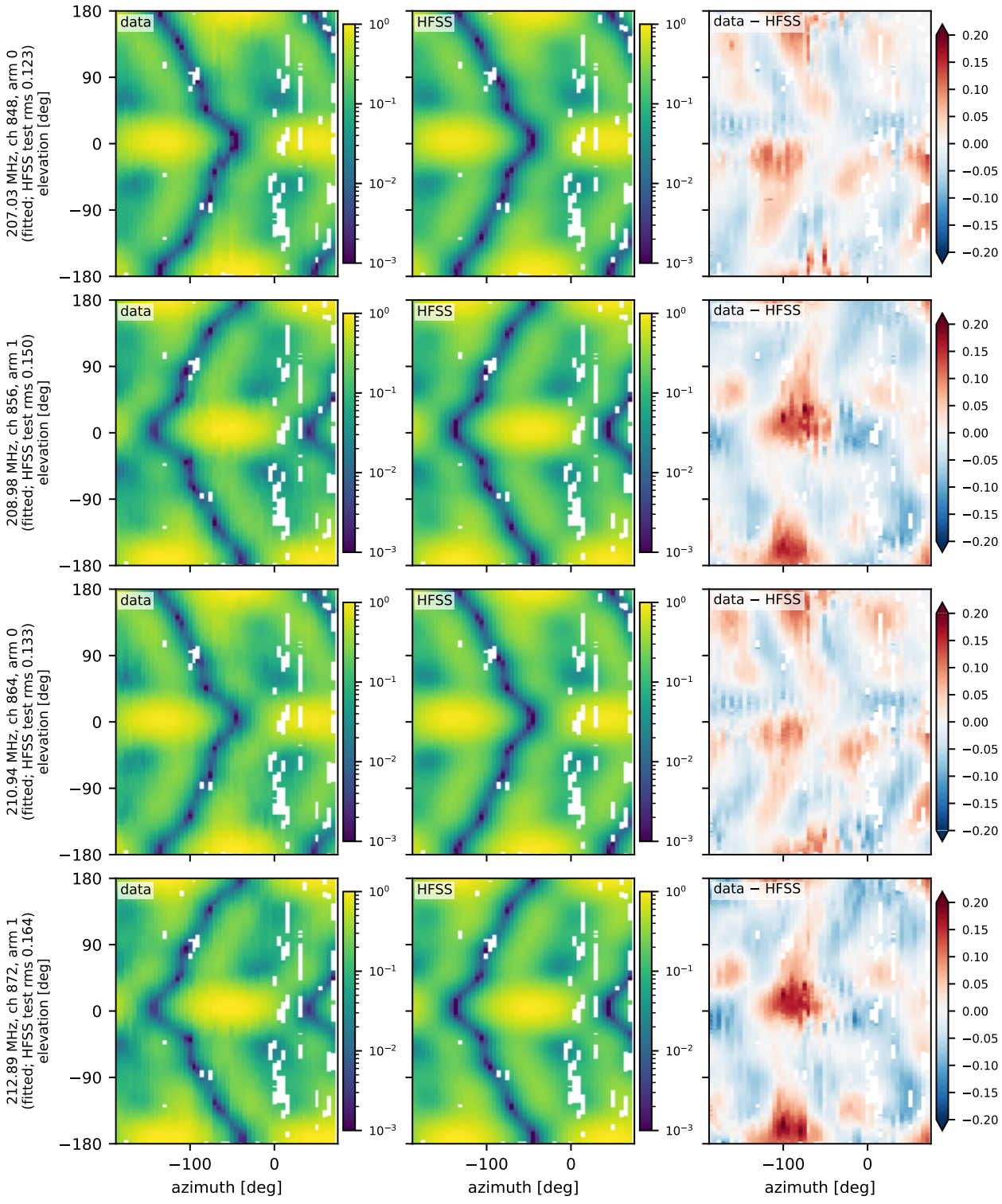


Figure 28: Raster maps, 207.03–212.89 MHz (page 21 of 26); panels as in Fig. 4.

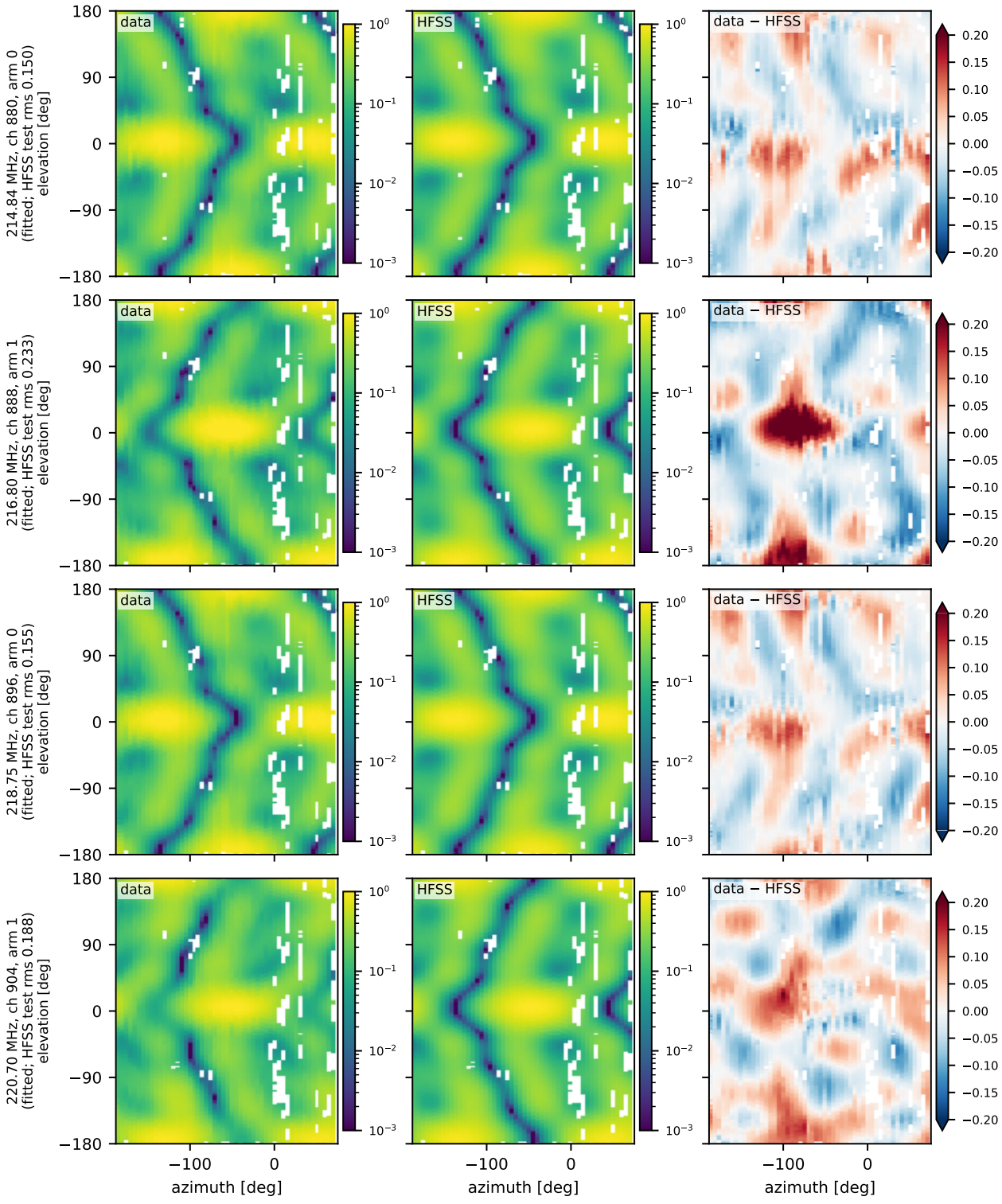


Figure 29: Raster maps, 214.84–220.70 MHz (page 22 of 26); panels as in Fig. 4.

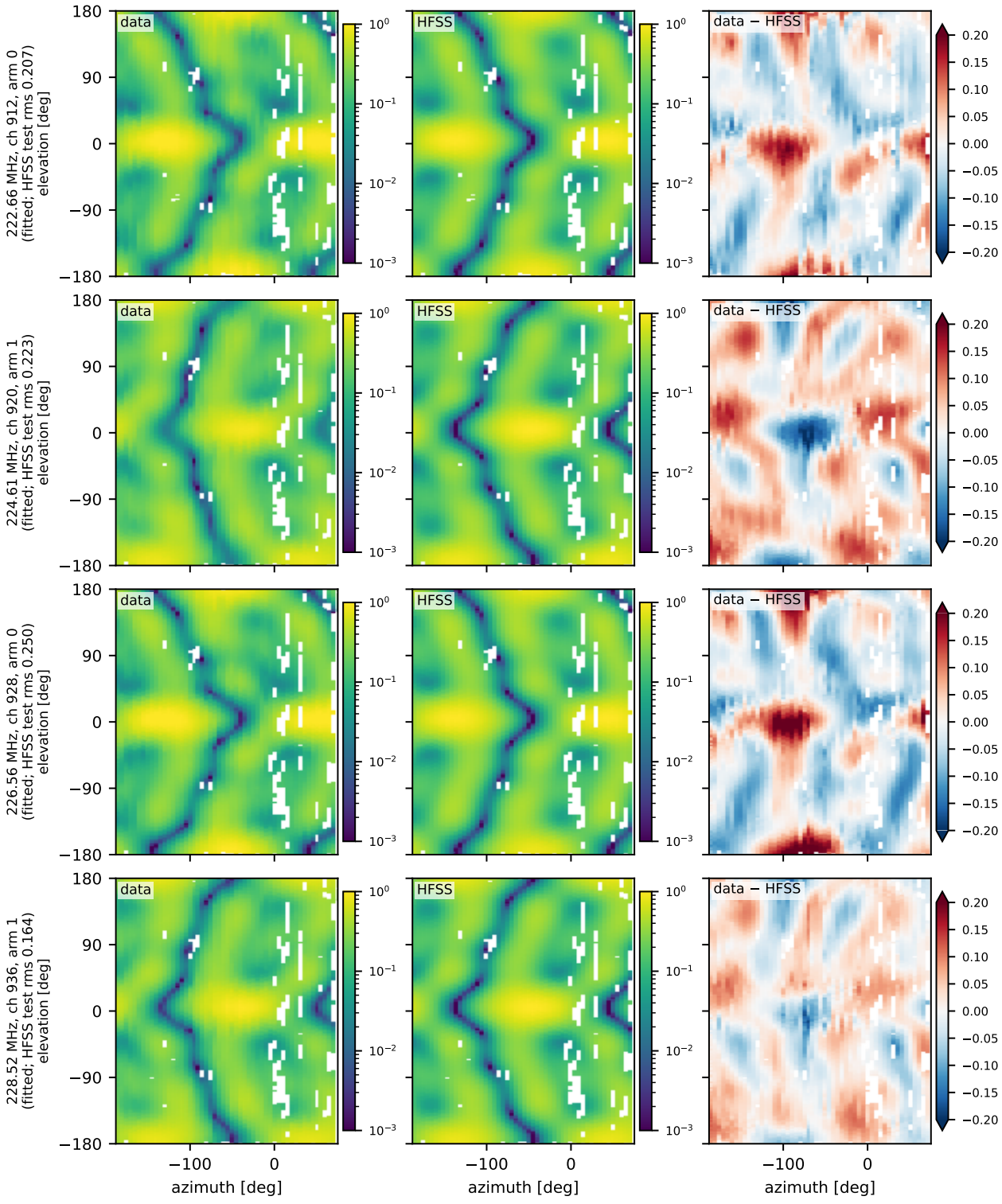


Figure 30: Raster maps, 222.66–228.52 MHz (page 23 of 26); panels as in Fig. 4.

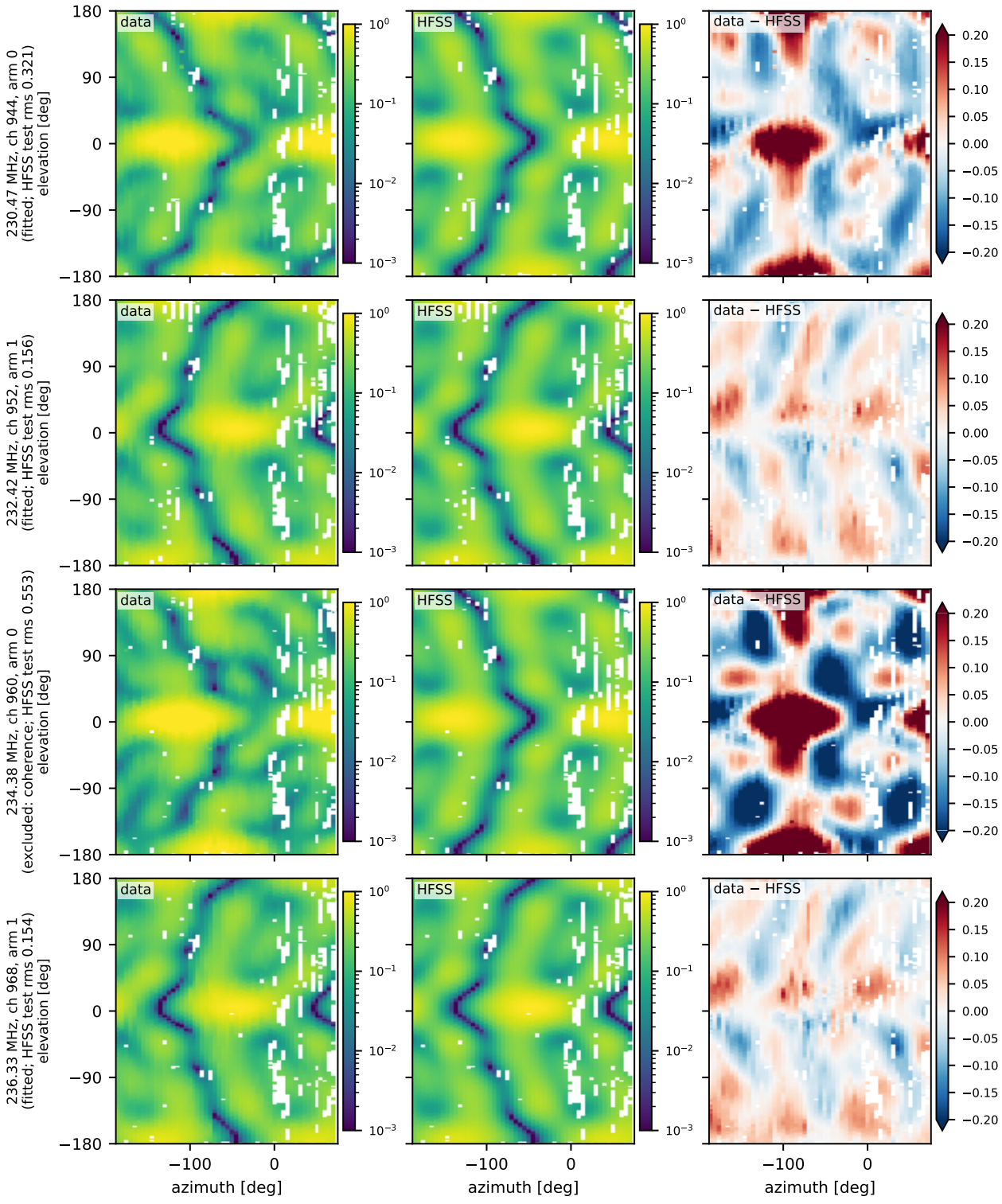


Figure 31: Raster maps, 230.47–236.33 MHz (page 24 of 26); panels as in Fig. 4.

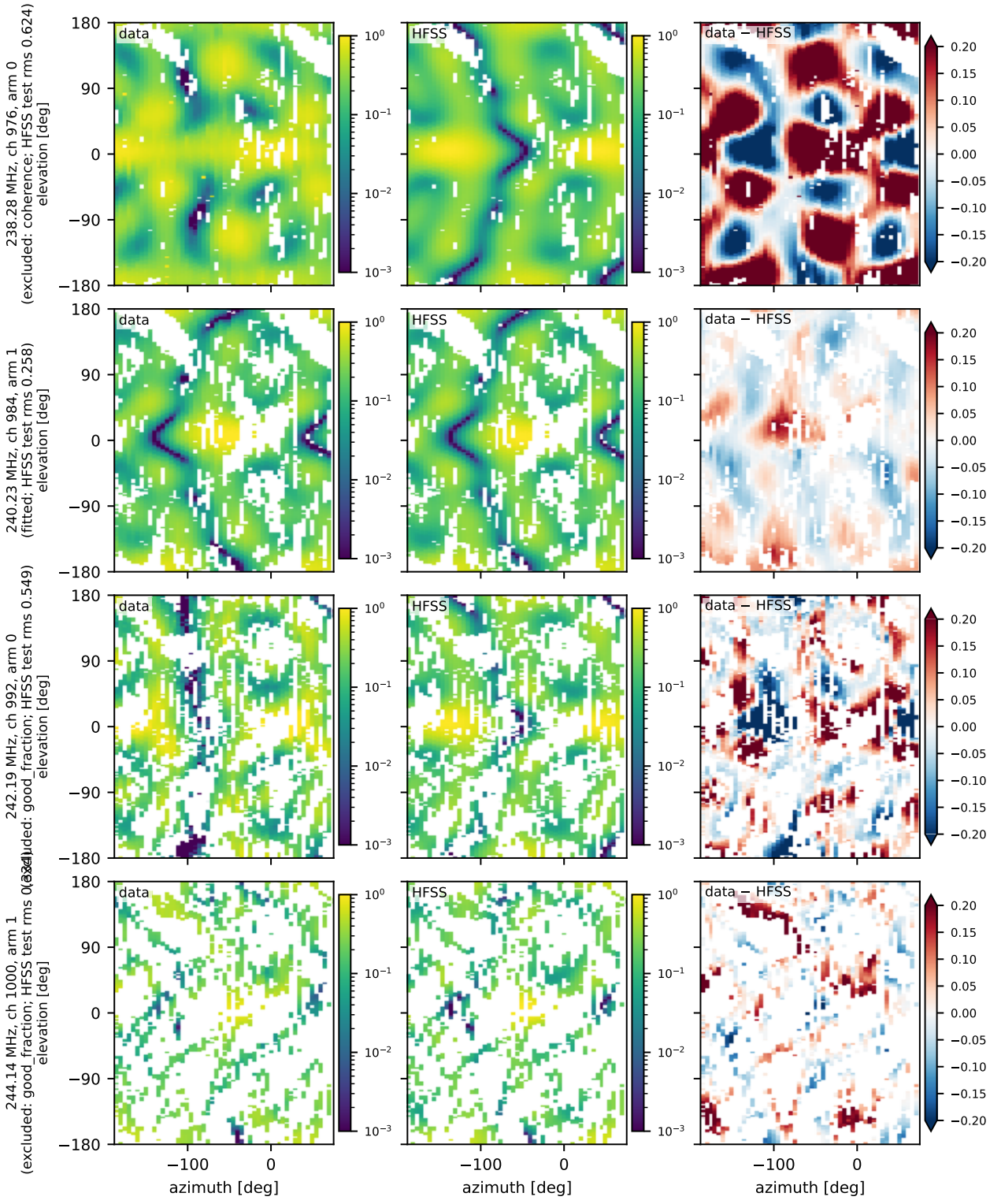


Figure 32: Raster maps, 238.28–244.14 MHz (page 25 of 26); panels as in Fig. 4.

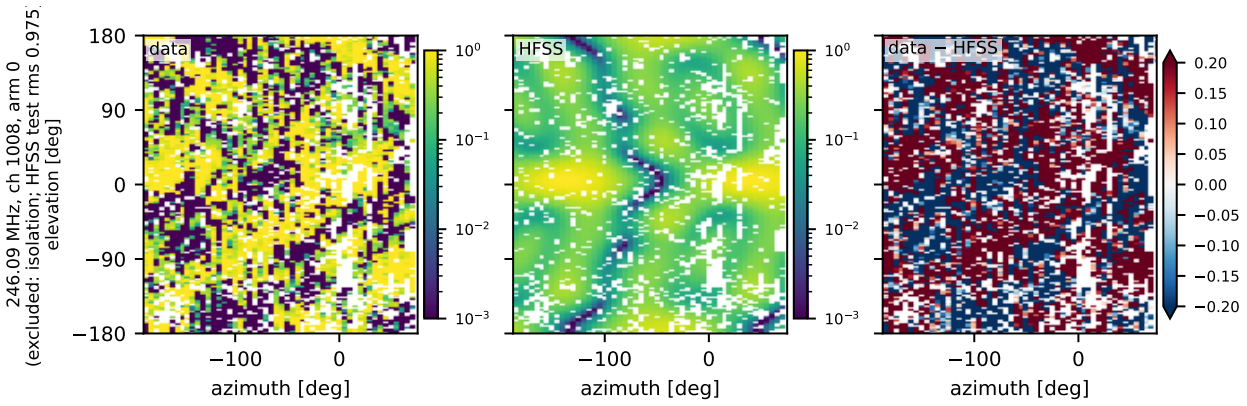


Figure 33: Raster maps, 246.09 MHz (page 26 of 26); panels as in Fig. 4.